

When less is more: The puzzling impact of employee education on green innovation performance

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ABSTRACT

Environmental sustainability increasingly requires that firms adopt green innovation (GI) to balance their strategy for ecological protection with financial performance. However, the role of employee education in shaping this relationship remains ambivalent. Drawing on the World Bank Enterprise Survey, which covers 8941 firms across 42 countries, we employ a two-stage least squares approach to estimate these effects. Results show that GI improves overall financial performance, though the effect is insignificant in knowledge-intensive business services and among Asian firms. Strikingly, we uncover a “less is more” dynamic that challenges proportionality assumptions in human capital theory: GI's financial performance benefits are strongest at low levels of formal employee education and decline to negligible effects as education increases. This pattern is most pronounced in low-tech manufacturing, among European firms, and in large organizations. Extending absorptive capacity theory to sustainability, this suggests that practical, adaptive skills outweigh theoretical knowledge in translating GI into performance gains. We recommend that firms balance formal education with experiential learning and hands-on implementation skills to enhance GI performance.

1. Introduction

Environmental impacts driven by economic activities are increasingly alarming and have concurrently triggered an urgent need for actions that support sustainability paradigm shifts. Projects, such as the European Green Deal, have been introduced to address these new environmental realities (IPCC, 2023), given the need for inclusive ecological protection. The focus is primarily on increased resource efficiency and on new avenues for social development that do not exacerbate environmental degradation (European Commission, 2011; Sala et al., 2014). In the current context, green innovation (GI) has become a transformative business strategy that addresses some environmental challenges and meets the growing consumer demand for sustainable solutions (Pinkse, 2023). Business analysts argue that companies that are leaders in sustainable practices and environmentally sound green technologies have a crucial competitive edge (Bentham, 2024). Lower operational costs, larger market share, increased brand equity, and improved corporate perception are the means through which this competitive edge has been recognized (Winston, 2024). Consistent with the resource-based view (RBV) (Barney, 1991), these economic gains

illustrate how GI capabilities create sustainable competitive advantages (Andersén, 2021; Silvério et al., 2025).

In this regard, GI becomes imperative for firms seeking to reconcile environmental responsibility with competitive advantages, such as financial performance. Here, there is an emergence of “green gazelles” whereby firms can achieve rapid growth by deploying technologically and regulatory-responsive strategies to facilitate eco-driven competitiveness (Colombelli et al., 2015). These firms can align with environmental requirements to catalyze strategic differentiation, particularly because they possess absorptive capacity to identify, assimilate, and apply external knowledge (Cohen and Levinthal, 1990; Silvério et al., 2025; Zahra and George, 2002). From the RBV perspective, GI capabilities represent how organizations utilize human capital and knowledge assets to create unique value in sustainability-driven markets, such as securing long-term economic benefits (Becker, 1975; Crook et al., 2011). In this context, human capital, such as employee education, is often considered an enabler of innovation. Prior studies indicate that educated and skilled workers are fundamental in facilitating environmental knowledge and turning it into profitable innovation (Barba-Aragón and Jiménez-Jiménez, 2024; Shafiullah et al., 2021;

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Wang et al., 2022). Nonetheless, despite this conception, the influence of employee education on shaping GI performance remains underexplored and potentially paradoxical.

Consequently, the traditional premises of human capital theory and the RBV may not fully capture the dynamics of GI implementation. From the perspective of absorptive capacity theory (ACT), employee education generates stratified absorptive capacity, whereby different levels of educational attainment contribute distinct yet complementary capabilities (Aghion et al., 2009; Cohen and Levinthal, 1990; Leal-Rodríguez et al., 2013). We challenge the assumption that absorptive capacity uniformly enhances GI returns across all education levels and instead propose that higher education levels may reduce the marginal financial benefit of GI. Highly educated workforces possess greater theoretical absorptive capacity. However, they may experience diminishing or even negative returns to GI when formal knowledge structures misalign with the practical, incremental nature of many GIs. This framing situates our study to explore how education moderates GI effectiveness, extending ACT to the context of sustainability transitions. We propose that education's effect on GI performance is context-dependent and varies systematically across firms. It is shaped by educational distribution (workforce coordination patterns), contextual fit (alignment between knowledge types and GI demands), and cognitive load constraints (implementation pragmatism versus analytical complexity). Lower and moderate levels of education may generate stronger GI returns when implementation agility and operational integration outweigh purely research-intensive capabilities, particularly in manufacturing contexts where GIs are predominantly incremental and process-oriented. Following Wales et al. (2013), organizational capabilities are most effective when aligned with contextualized contingencies, so the strategic value of employee education depends on its fit with organizational structure, sectoral demands, and institutional environments. This perspective challenges traditional RBV assumptions about linear human capital optimization and suggests that increasing workforce education can reduce the marginal financial benefit of GI in certain contexts. To empirically test this proposition, we analyze data from the World Bank Enterprise Surveys (WBES) on 8941 firms across 42 countries using a two-stage least squares (2SLS) approach. We advance this conversation by unpacking how employee education moderates the relationship between GI and financial performance, while accounting for firm heterogeneity across contexts and firm characteristics.

Our investigation addresses three core questions: Does employee education enhance GI's financial effectiveness? How do education levels shape this relationship across contexts? Moreover, to what extent does the moderating effect of education vary by firm characteristics? While we confirm that GI improves financial performance (Shuwaikh et al., 2023), this effect is conditioned by organizational and contextual heterogeneity. Moreover, we provide insights that qualify dominant assumptions about human capital optimization. We question the presumption of a linear positive relationship between employee education levels and GI outcomes (Domadenik et al., 2020; Consoli et al., 2016). We uncover a counterintuitive “less is more” dynamic: firms with lower levels of education often derive greater performance benefits from GI than those with higher levels of education. This suggests that employees' absorptive capacity is not uniformly enhanced by education, but instead shaped by its distribution, contextual fit, and cognitive load constraints. By bridging gaps in the literature and integrating insights from RBV, the ACT, and sustainability transitions theory (STT), our study offers a differentiated understanding of how human capital configurations influence the financial performance of GI. Overall, the findings offer practical implications for workforce development, policy plans, and strategic management of sustainability-driven innovations.

The remaining part of the paper is configured as follows: In Section 2, we present the theoretical perspectives and the tested hypotheses. Next, we described the research method in Section 3, followed by the empirical analysis and results in Section 4. These results are discussed in Section 5, including the research implications, and the paper concludes

in Section 6.

2. Theoretical perspectives and research hypotheses

This section reviews existing research on the relationship between GI and firm financial performance, with a focus on the role of employee education. It critiques traditional human capital and RBV perspectives, which often assume a linear link between education and innovation capacity. Drawing on ACT and complemented by STT, the section introduces the concept of stratified absorptive capacity, proposing that different education levels contribute distinct yet complementary capabilities for acquiring and applying green knowledge.

2.1. Green innovation and financial performance

Sustainability theory provides a basis for conceptualizing an inter-connectivity of environmental protection, economic growth, and social well-being (Purvis et al., 2018). It conceptualizes sustainability as a trade-off among the pillars and a process of system change to resilient socio-technical systems that ensure intergenerational equity. Transition theory extends this view by emphasizing the co-evolution of technology, institutions, policies, and culture required to set economies on low-carbon and resource-efficient trajectories (Hundloe, 2021). Transitions are multilevel processes of reconfiguration wherein innovation, governance, and stakeholder collaboration interact to yield long-term systemic change (Hundloe, 2021; Purvis et al., 2018; Zhai and Chang, 2018). The RBV paradigm complements sustainability strategies, given that it explains how “valuable,” resource-based competencies that are difficult to imitate enable businesses to achieve a competitive advantage (Barney, 1991). This model is further extended to argue that GI, as a dynamic capability, enables companies to turn environmental pressures into strategic opportunities for a competitive edge (Andersén, 2021; Silvério et al., 2025).

Empirical studies progressively position GI as a driver of firm-level competitiveness and broad sustainability transitions. According to Colombelli et al. (2015), eco-innovation is one of the principal accelerators of the transition to a green economic path. The study confirmed that only those firms that follow an innovation-based strategy are likely to present stronger growth and environmental performance. This is expected in light of the technological-innovation-systems perspective that sustains the theory of sustainability transitions. In line with this notion, evidence from Jia et al. (2021) supports the view that green technology innovation is a means of achieving sustainable industrial practices. Behavioral and organizational factors shape firms' capacity to adapt to sustainability transitions. These transformations are also shaped by policy and institutional contexts (Leoncini et al., 2019). As noted by Holzner and Wagner (2022) and Acemoglu et al. (2023), accelerating the transformation toward eco-friendly markets is possible through the establishment of appropriate policy instruments, economic incentives, and governance mechanisms. These tools are developed to align markets with ecological targets and stimulate the adoption of green technologies. This conforms with the multiple levels of analysis in the literature on sustainability transitions, whereby innovations at the niche level, such as GI, interact with higher-level regimes, such as policy, markets, and society, which regulate unsustainable systems.

Such a context enables GI to establish a linkage among environmental resilience, technological development, and economic performance (Purvis et al., 2018; Zhai and Chang, 2018). From the perspective of RBV, it reconfigures organizational assets to simultaneously pursue both environmental and economic goals (Andersén, 2021; Silvério et al., 2025). Empirical evidence supports these perspectives: Colombelli et al. (2015) and Holzner and Wagner (2022) show that GI improves environmental efficiency and economic resilience, while Acemoglu et al. (2023) show that it plays a role in developing the clean economy through purposefully deployed policy and technology channels. Shuwaikh et al. (2023) provide more evidence that GI investments have

both financial and environmental benefits, while [Jia et al. \(2021\)](#) indicate increased cost efficiency and profitability in places with strong sustainable development policies. [Dangelico \(2016\)](#) connects green product innovation with cost advantage and market differentiation, and [Banelienė and Strazdas \(2023\)](#) show a positive relationship between GDP growth and GI in the European Union. Reconciling these findings indicates that GI is both a firm capability and an institutional mechanism that connects corporate performance to national sustainability objectives ([Leoncini et al., 2019](#)).

Refinement with open innovation and environmental Kuznets perspectives further deepens this observation. [Shearmur and Doloreux \(2016\)](#) reveal that innovation processes vary spatially according to network density, market reactions, and knowledge exchange—meaning that open innovation ecosystems enhance business organizations' capacity to create and spread green technologies. GIs achieve system-level impacts through the combined influence of external environments and organizational partnerships within sustainability transitions. In addition, [Shafiullah et al. \(2021\)](#) extended the Environmental Kuznets Curve and identified human capital and education as drivers of a stronger linkage between economic paradigms and the environment. However, Shafiullah et al. called for public learning capacity and awareness as institutional and social levers of change in the long run. It is these insights that led to the intersection of these perspectives, proposing a multidimensional convergence of institutional, organizational, and technological factors on GI. Thus, by connecting the transition framework, the RBV, and sustainability theory, the GI becomes a strategic capability that not only improves firms' financial performance but also ensures long-lasting organizational survival. Within this integrated frame, we present our first hypothesis:

Hypothesis 1. A firm's GI efforts will positively affect its financial performance in a sustainability-driven economy across contexts.

2.2. Green innovation, financial performance, and employee education link

The RBV theory posits that organizational resources drive sustainable performance through strategically developing, combining, and deploying unique organizational resources ([Grant, 1991](#)). Human capital, as a core intangible asset, becomes central to firms' ability to adapt and thrive in sustainability-driven economies. Within the integrated framework of sustainability transitions, employee education enhances firms' dynamic capabilities by enabling them to reconfigure internal competencies in response to environmental pressures ([Barney, 1991](#); [Teece, 2016](#); [Wernerfelt, 2016](#)). The ACT complements this view by explaining how firms leverage employee education to recognize, assimilate, and apply external knowledge for GI ([Cohen and Levinthal, 1990](#)). In this context, education is not merely a resource but a strategic enabler of GI that aligns organizational learning with broader socio-technical transitions.

Employee education plays a fundamental role in the firm's absorptive capacity development (potential – acquisition/assimilation and realized – transformation/exploitation) ([Zahra and George, 2002](#)). Such capabilities are indispensable for addressing the multilevel reconfigurations that STT demands, in which firms interact with emerging policy, market, and cultural regimes ([Hundloe, 2021](#)). However, absorptive capacity operates within cognitive constraints that vary by educational stratification. Organizational members experience differential cognitive load based on their academic background and the complexity of knowledge encountered ([Leppink and Van den Heuvel, 2015](#)). Within the context of GI, Education also provides employees with critical thinking and interpretive skills. These skills enable them to recognize and apply environmental technologies, expanding firms' latitude to comply with environmental imperatives or policy changes ([Acemoglu et al., 2023](#); [Schweisfurth and Raasch, 2018](#)).

Previous studies have shown that employee education and training,

as well as employee engagement, are essential for the positive impact of GI capabilities on financial performance ([Barba-Aragón and Jiménez-Jiménez, 2024](#); [Liao, 2017](#)). Green human resource management activities ([Aloqaily, 2023](#)) and corporate social responsibility towards employees ([Guinot et al., 2023](#)) facilitate the creation of inclusive innovation cultures that enhance human capital, leading to overall improvement. Critically, an organization's absorptive capacity depends on an educated workforce and the spread of knowledge across hierarchical and functional layers. This view is consistent with the emphasis on stakeholder co-operation and institutional alignment in sustainability transitions, in which organizations must manage the tension between depth of competence and adaptive capability across precedence levels ([Leoncini et al., 2019](#); [Rho et al., 2021](#)). The incorporation of GI into wider transition pathways demands not only technical skills but also the capacity to engage with institutional complexity and market forces.

However, ACT reveals a paradox: while employee education enhances innovation capacity, its contribution to GI may diminish as education levels increase. This perception reflects stratified cognitive constraints whereby highly educated employees, particularly when concentrated in specialized functions, may experience cognitive overload. Such overload can hinder the flexible problem-solving and cross-functional coordination required for effective GI deployment ([Bunjak et al., 2021](#); [Stajkovic and Stajkovic, 2025](#); [Wales et al., 2013](#)). From a “less is more” perspective, GI outcomes may be stronger in firms with lower or more balanced educational compositions, where adaptability and implementation pragmatism outweigh analytical complexity ([Leal-Rodríguez et al., 2013](#)). This logic suggests that excessive concentration of advanced education—without complementary practical and mid-level expertise—can create cognitive silos that impede innovation ([Aghion et al., 2009](#)) and reduce responsiveness to sustainability transitions ([Yuwono, 2021](#)).

The institutional and infrastructural framework that enterprises use to build their absorptive capacity is shaped by regional innovation ecosystems ([Asheim and Gertler, 2005](#)). These ecosystems differ markedly by geography, affecting the way firms utilize educated employees for GI. In developed economies, companies leverage strong policy backing, network knowledge, and market stimuli to consolidate financial advantages from GI ([Colombelli et al., 2015](#); [Crescenzi and Rodríguez-Pose, 2011](#); [Jia et al., 2021](#)). In contrast, firms in developing contexts may face institutional voids that limit the effectiveness of highly educated personnel unless complemented by strategic leadership and organizational culture ([Putro and Lubis, 2023](#)). Thus, employee education shapes the financial impact of GI in a context-dependent and diminishing manner ([Aghion et al., 2009](#)). Even though green jobs require advanced cognitive and social skills ([Consoli et al., 2016](#)), excessive formal education may lead to disproportionate confidence or rigidity, especially when institutional complementarity is missing. To address this, firms should strategically invest in green human capital to ensure that capability enhancement does not hinder dynamic problem-solving ([Domadenik et al., 2020](#); [Shafiullah et al., 2021](#); [Wang et al., 2022](#)). Structuring the workforce across educational tiers helps mitigate cognitive overload and insular thinking. Evidence suggests that excessive formal education does not necessarily translate into superior GI outcomes, particularly when implementation relies on incremental and process-oriented changes ([Buhl et al., 2016](#); [Riaz et al., 2024](#)). Building on this, we argue that the financial impact of GI depends on employees' educational levels. While education enhances absorptive capacity, its interaction with GI is subject to cognitive constraints and diminishing returns at higher levels. Accordingly, we expect a negative moderating effect, with the financial benefits of GI strongest at lower levels of employee education and weakening as education increases. Hence, we hypothesize that:

Hypothesis 2. The positive financial performance impact of GI is stronger at lower levels of employee education than at higher levels.

3. Methods

This section describes the empirical approach used to test the study's hypotheses, using key variables, including GI adoption, financial performance, and employee education, along with relevant control variables. It addresses potential endogeneity in the relationship between performance and GI, employing a 2SLS estimation strategy.

3.1. Data and sources

This research draws upon the 2018-2019 WBES, which collects firm-level data from the global private sector. The dataset encompasses various industries, including most manufacturing sectors and selected retail, wholesale, hospitality, and construction services, excluding agriculture, mining, banking, and finance firms. The WBES uses a standardized questionnaire and a consistent stratified sampling method that accounts for a country's geographic region, industry, and size to represent each country's private economy accurately. We use an aggregate sample of 8941 cross-sectional firm observations drawn from 42 countries.

3.2. Variables

3.2.1. Financial performance and green innovation

Our dependent variable, financial performance, is proxied by two financial metrics—sales and profit growth—to distinguish GI's effects on market performance from net financial performance outcomes. This dual-outcome approach aligns with prior studies examining environmental innovation's financial impacts, enabling the capture of distinct mechanisms through which green practices affect firm performance (Le, 2022; Zhang et al., 2019). Sales growth is expressed as the proportional change in sales over the past three years, normalized by the current year's sales. Aligning with *FasterCapital* (2024), profit growth is determined by multiplying sales growth by the GPM (defined in Table 1). The GPM is calculated as total sales minus the total cost of goods divided by total sales.

GI is the primary covariate expected to affect financial performance. The concept of GI has been interpreted differently across the academic domain. However, we evaluate GI by examining firms' energy consumption patterns that follow specific measures that meet green practices. The categories include enhancing heating and cooling systems, implementing on-site climate-friendly energy generation, upgrading machinery and equipment, adopting energy management systems, modernizing vehicle fleets, improving lighting systems, installing air pollution control technologies, improving water management, minimizing waste through recycling and waste management, and implementing other pollution control measures. In the 2019 WBES survey,

Table 1
Adoption Rates for Eco-Friendly Investment Categories (N = 8941 firms).

Category	Adoption rate among firms (%)
Heating and cooling improvements	35.97%
More climate-friendly energy generation on-site	14.74%
Machinery and equipment upgrades	46.85%
Energy management	32.35%
Waste minimization, recycling, and waste management	39.38%
Air pollution control measures	15.88%
Water management	21.61%
Upgrades of vehicles	33.45%
Improvements to lighting systems	46.45%
Other pollution control measures	12.17%

Note: The table reports the share of firms that adopted each eco-friendly investment category during the three years preceding the survey. Firms may report adoption in multiple categories; therefore, percentages do not sum to 100 percent.

firms were asked whether they had undertaken any of the ten eco-friendly investments over the past three years. In the survey, a firm is classified as having implemented GI if it reported adopting at least one of these measures within that period. For each firm, we computed a raw GI count by summing binary indicators of adoption across the 10 categories, yielding values ranging from 0 (no GI) to 10 (firm-level adoption of all measures). Table 1 shows that adoption levels are concentrated at the lower end, suggesting that most firms engage in limited green upgrading.

That said, we computed sector-specific benchmarks for the GI count to normalize firm-level variation. For each sector, we calculated the mean ($\bar{GI}_s$) and standard deviation (σ_s) of firms' raw GI counts. Each firm's standardized GI score was obtained as:

$$GI_{is}^{std} = \frac{GI_{is} - \bar{GI}_s}{\sigma_s}, \quad (1)$$

where mean GI_{is} denotes the raw GI count of firm i in sector s . This transformation expresses a firm's GI activity relative to its sector's mean, allowing for meaningful cross-sector comparability and mitigating industry-level heterogeneity in green practices. We verified that no sector exhibited zero variance ($\sigma_s = 0$), ensuring valid standardization. The resulting standardized GI score serves as the primary covariate in the empirical models examining the association between green investment and firm financial performance.

Fig. 1 shows a right-skewed distribution that confirms Table 3, indicating that most firms adopt only a few green practices, while only a minority adopt more. Supplementary Information: Table 5 describes how GI scores vary across geographic regions and firm sectors. Scores are highest in North Africa and Asia, where they align strongly with sales and profit growth. European firms show no significant link. In manufacturing, GI shows positive sales growth but not profit growth. In the service sector, trends are weak and mixed.

3.2.2. Other determinants of financial performance

From a knowledge-based perspective, we control for multiple determinants when modeling GI's impact on a firm's financial performance. Employee education is one of the closely aligned factors in enhancing firms' adaptive capacity. Consoli et al. (2016) show that educated employees strongly support driving green occupations and optimally sustaining sustainability practices. Thus, higher levels of education most likely provide skills and knowledge relating to the understanding and application of environmental innovations capacity. Thus, in sum, these studies suggest that a more educated workforce can help build the knowledge and learning capabilities required for effective GI. To empirically capture this perception, we constructed an education index that reflects a firm's workforce's overall educational attainment. The index uses two key indicators: the proportion of full-time permanent employees who completed secondary school (edu 1) and the proportion who hold a university degree (edu 2). Specifically, building on the preceding background, we assigned a weight of 0.5 to secondary education and a full weight of 1.0 to university-level education. The index was developed as follows:

$$Education\ index = \frac{0.5(edu1) + 1.0(edu2)}{100} \quad (2)$$

By incorporating this index into our analysis, we anticipate the extent to which employee education affects financial performance and moderates the impact of GI. The index ranges from 0 to 1.5, with higher values indicating a more educated workforce.

Besides human capital, market competition significantly influences increased revenue. Accordingly, firms with monopoly power can boost sales through pricing strategies, while intense competition typically compresses margins (Aghion, 2018). Industry effect is also included, as concentration level is vital in determining a firm's increasing revenue (Aghion et al., 2005; Eide et al., 2021). As export orientation correlates

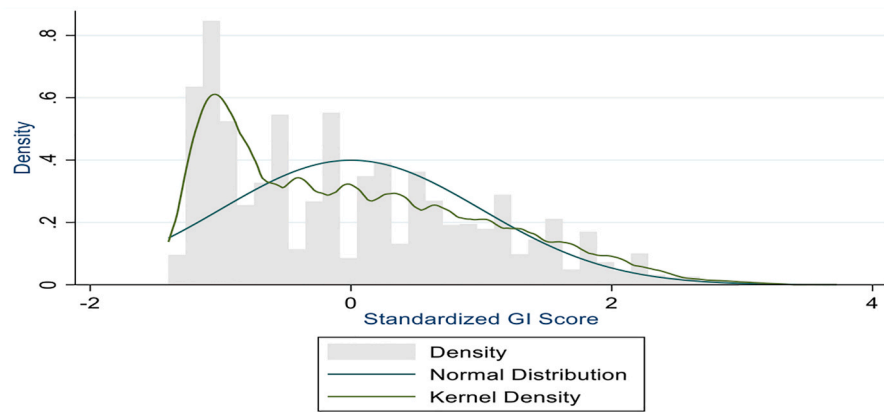


Fig. 1. Distribution of standardized GI score.

Note: The histogram shows the frequency distribution of standardized green innovation scores across the sample of firms. The blue line represents a normal distribution overlay, while the green line shows the kernel density estimate. GI = Green Innovation.

with higher productivity and potential returns (Antonelli, 2019), we included export status to examine how export activities affect profitability. We include the legal structure of firms because corporate forms influence profits through enhanced capital access and operational flexibility (Stowers, 2024). Also, we include firm age, although previous research indicates it exhibits mixed effects on financial performance. Some studies indicate a negative relationship (Akben-Selcuk, 2016), and others suggest positive impacts (Matemilola et al., 2024). Additionally, we include foreign ownership of businesses as it positively correlates with firm value and performance, as evidenced in multiple studies (Choi and Park, 2019; Zhang, 2023).

Further, we account for a firm's geographic location, including regional and development factors, as well as the city in which it operates. Considering these additional factors, we seek to isolate the core relationships of interest to minimize the risk of omitted-variable bias. Geographic fixed effects can further help explain how GI influences financial performance, accounting for regulatory environments, economic stability, and market size (Nahm, 2022). In this context, Europe offers an instance of a mature innovation ecosystem with well-established institutions, knowledge infrastructures, and inter-firm networks (Asheim and Gertler, 2005). Asia and North Africa represent developing innovation ecosystems where institutional support and knowledge infrastructure are still evolving, potentially creating different dynamics for how employee education translates into GI performance.

In what ways sectoral differences (manufacturing vs services) influence financial performance is also considered. The reason is that industries exhibit distinct operational characteristics and capacities that may affect their ability to implement and benefit from GI (Porter and Linde, 1995). Firm size is another critical factor. More robust financial resources are linked with larger firms' need to support research and development (R&D) investments and manage risks associated with innovation (Cohen and Klepper, 1992; Knott and Vieregger, 2015; Leal-Rodríguez et al., 2013). Large firms also gain an advantage from economies of scale and scope, as it enables them to spread costs across their production processes, enhancing their capacity to innovate (Martínez-Costa et al., 2014). In this context, these types of firms may be better equipped to engage in GI than smaller firms. Even though resources are more limited in smaller firms, they may capitalize on their agility and niche-market strategies to drive proportional sales growth and thereby gain market power (Sen et al., 2023). In this regard, financial performance may be more apparent in small and medium-sized enterprises (SMEs) than in larger firms, depending on their business location or region (Gaio and Henriques, 2018). To control for this dynamic, we used SMEs as the base category and large firms as the comparison group.

3.2.3. Descriptive statistics

Table 2 shows how these variables were measured, and descriptive statistics for the variables analyzed across the whole sample are provided in Table 3. Most variables are binary, except for the financial performance metrics, GI, R&D, industry concentration, education index, and firm age, which are continuous. We utilize the inverse hyperbolic sine (IHS) transformation for variables that include negative and zero values. Unlike the natural logarithm, it is defined across the entire real line, allowing us to retain all observations and preserve the sign and relative magnitude of the original data (Aihounton and Henningsen, 2021). Supplementary Information Table S1 to Table S3 detailed how the sample was broken down. It consists of 5155 manufacturing and 3786 service firms for sectoral analysis. 1606 North African firms, 5504 European firms, and 1831 Asian firms for regional analysis. Also, it includes 6213 SMEs and 2728 large firms for size analysis. Consequently, Supplementary Information Table S4 provides further details on the 42 countries studied, grouped by geographic regions.

3.2.4. Instrument variable

Because firms with stronger financial performance are more likely to invest in GI, endogeneity caused by reverse causality arises. To address this concern, we draw upon an IV strategy (Clougherty et al., 2016). Prior research has often used R&D expenditure as an instrument for innovation, as it reflects firms' technological capabilities and innovative potential (Bogliacino and Pianta, 2013). However, firm-level R&D can also directly influence financial performance outcomes through non-GI, process optimization, or brand development, potentially violating the exclusion restriction. To mitigate this issue, we construct a region-standardized R&D intensity instrument (rd_std). For each region, we calculate the mean of firm-level R&D and standardize each firm's value accordingly. This measure captures a firm's relative R&D exposure within its regional context and serves as a strong predictor of GI. By including region fixed effects, the instrument relies solely on within-region variation, which is less likely to affect financial performance, except through GI directly. This approach aligns with prior evidence that regional research intensity and spillovers shape firms' innovation behavior (Jaffe, 1989) and that regional R&D variation can serve as an exogenous determinant of innovative activity (Bronzini and Piselli, 2016).

3.3. Model specification

Besides endogeneity arising from reverse causality, omitted factors, such as managerial ability, organizational culture, and strategic orientation, jointly influencing both GI decisions and financial performance outcomes can bias results. Hence, the relationship between GI and firm

Table 2

Measurement details of all variables.

Variable	Definitions of Variables
Gross Profit Margins (GPM)	GPM is calculated as total sales minus total cost of goods sold, divided by total sales.
Sales growth	Expressed as the proportional change in sales over the past three years, normalized by the current year's sales.
Profit growth	Profit growth is computed as the product of sales growth and GPM. This computation is also known as Profit Contribution Growth (FasterCapital, 2024).
Green Innovation (GI)	GI is computed as a sector-standardized index based on the number of green measures a firm adopted (0-10), obtained by standardizing each firm's raw adoption count relative to its sector's mean and standard deviation.
Green innovators	Computed as managerial experience, measured relative to the regional mean (=1 and 0 otherwise).
Education index	The education index is a weighted composite measure of a firm's workforce education, calculated as $0.5 \times$ (the share of employees with secondary education) + $1.0 \times$ (the share with university education), yielding values from 0 to 1.5 that reflect overall educational attainment.
Exporter	A dummy variable (=1 for exporting firms and 0 otherwise)
Firm legal status	A firm conducting business under the legal arrangement of corporate firms (=1 and 0 otherwise)
Firm age	The age of the firm in years, on a logarithmic scale
Foreign ownership	Firms partially owned by foreign investors (=1 and 0 indicates otherwise)
SMEs	Small and medium firms encompassing 5 to 99 employees based on the World Bank classification (=1 and 0 indicates otherwise)
Large firms	Firms with 100 or more employees (=1 and 0 indicates otherwise)
Small City	Firms located in areas with a population of less than 50,000 (=1 and 0 indicates otherwise)
Medium-sized city	Firms situated in areas with a population of around 50,000 (=1 and 0 indicates otherwise)
Govt. spending on education	The government should prioritize spending on education (=1 and 0 indicate otherwise)
Research and Development	Research and development is measured as a region-standardized intensity variable, obtained by normalizing each firm's R&D expenditure relative to the regional mean to capture its R&D effort within the local innovation context.
Manufacturing firms	It corresponds to firms classified with ISIC codes 15-37
Hightech_firms	It corresponds to firms classified with ISIC codes ≥ 23 & ≤ 35
Low-tech firms	It corresponds to firms classified with ISIC codes ≥ 15 & ≤ 22 plus ISIC codes ≥ 36 & ≤ 37
Service firms	It corresponds to firms classified with ISIC codes 49-59
KIBS firms	It corresponds to firms classified with ISIC codes 60 -72
Region dummies	We included Asia, North Africa, and Europe (Each region = 1 when others = 0)

Source: WBES: <https://login.enterprisesurveys.org/content/sites/financeandprivatesector/en/signin.html>.

financial performance is endogenous. To address these endogeneity concerns, we employ a 2SLS approach. Following Wooldridge (2010), our model is formally specified as:

$$y_i = GI_i\beta_1 + \mathbf{x}_{1i}\beta_2 + u_i \quad (3)$$

$$GI_i = \mathbf{x}_{1i}\alpha_1 + \mathbf{x}_{2i}\alpha_2 + v_i \quad (4)$$

Where y_i represents financial performance (sales or profit growth) for firm i , and GI_i is an endogenous variable affecting y_i . $\mathbf{x}_{1i}$ (which includes GI, education index, export status, sector dummies ...) and $\mathbf{x}_{2i}$ (i.e., rd_std - the instrument for GI) are included and excluded exogenous variables, and β_1 and β_2 represent their coefficients. As well, α_1 and α_2 are coefficients linking the included variables to the endogenous variable GI , and u_i and v_i are zero-mean error terms, with nonzero correlations.

Table 3

Descriptive Statistics for the Total Sample (N = 8941 firms).

Variables	Mean	SD	Min.	Max.
Dependent variables				
Sales growth	0.133	0.262	-0.781	0.862
Profit growth	0.084	0.202	-0.764	0.859
Gross Profit Margin	0.633	0.364	-0.048	0.999
Independent variables				
Green innovation	-0.012	0.572	-1.706	2.720
Education index	0.617	0.344	0	1.5
Industry concentration	4.798	1.910	0	12.121
Exporter	0.072	0.259	0	1
Firm legal status	0.167	0.373	0	1
Firm age	2.891	0.654	0.693	5.308
Foreign ownership	0.104	0.306	0	1
SMEs	0.695	0.460	0	1
Instrument variables				
Research and Dev. (R&D)	-0.000	1.000	-0.448	3.169
Other controls				
Small-sized city	0.268	0.443	0	1
Medium-sized city	0.223	0.416	0	1
Asia	0.205	0.404	0	1
North Africa	0.180	0.384	0	1
Manufacturing firms	0.577	0.494	0	1

3.4. Estimation procedure

Among $\mathbf{x}_1$, we included an interaction term (*Education index* $\times$ *GI*) to assess how employee education moderates the effect of *GI* on firm financial performance. This context aligns with Luo and Xie (2023) and Shafiullah et al. (2021) on how human capital and institutional factors can drive sustainable business practices. Meanwhile, an interaction term of this nature, involving an endogenous variable, becomes endogenous and must be instrumented (Wooldridge, 2010). In our context, rd_std serves as the instrument for *GI*. As such, we instrumented the endogenous term (*Education index* $\times$ *GI*) with the interaction between rd_std and the education index (*Education index* $\times$ rd_std). Using Stata's ivregress 2SLS command to fit the equation, we specify the dependent variable y_i and all the included exogenous variables. We modeled the endogenous specifications and their instruments as: (*GI*, *Education index* $\times$ *GI* = rd_std , *Education index* $\times$ rd_std). To address within-country correlation and avoid biased inference, we estimated our baseline model using a 2SLS estimator with country-clustered robust standard errors, following Moulton's (1990) recommendation.

To gain a more nuanced understanding of how *GI* drives systemic change, we further examine three heterogeneity dimensions: geographic variations, representing different levels of innovation ecosystem maturity and market conditions; sectoral differences (manufacturing vs. services), and firm size (see Section 4.3). Even though Boschma and Gianelle (2014) conceptualize regions primarily at the subnational level, they argue that spatial and contextual heterogeneity mediates the economic impact of innovation. That is, heterogeneity at the regional level in infrastructure, institutions, and spillovers across space demonstrates how innovation is translated into economic performance (Crescenzi and Rodríguez-Pose, 2011). Drawing from the view that these theoretical backgrounds can be applied to broader geographic differences, we moved the conversation from within subnational and European contexts to much larger continental regions. We consider North Africa, Asia, and Europe to investigate how they shape the relationship between *GI* and financial performance. Beyond the fixed effect of firm size in the baseline models, we aim to understand how their distinct characteristics moderate *GI*'s financial performance outcomes, offering valuable insights. Thus, we assessed a firm-size model (SMEs vs. large firms) of the relationship between *GI* and financial performance, while controlling for key firm characteristics.

Table 4

Regression results for the baseline analysis – OLS and 2SLS models.

Independent variables	OLS + interaction		2SLS without interaction		2SLS + interaction (baseline model)	
	Sales growth	Profit growth	Sales growth	Profit growth	Sales growth	Profit growth
GI	0.029** (0.012)	0.016 (0.010)	0.075*** (0.027)	0.049** (0.021)	0.111*** (0.033)	0.070** (0.027)
Education index	0.014 (0.011)	0.018* (0.010)	0.011 (0.013)	0.016 (0.012)	0.002 (0.011)	0.011 (0.011)
GI × Education index	0.014 (0.012)	0.011 (0.010)			−0.053* (0.031)	−0.030 (0.031)
Competition	−0.002 (0.007)	−0.004 (0.006)	0.003 (0.009)	−0.000 (0.008)	0.003 (0.009)	0.000 (0.008)
Industry concentration	−0.001 (0.002)	−0.001 (0.002)	−0.000 (0.002)	−0.001 (0.002)	−0.000 (0.002)	−0.001 (0.002)
Exporter	0.022** (0.011)	0.015** (0.007)	0.006 (0.011)	0.004 (0.009)	0.005 (0.011)	0.003 (0.009)
Firm legal status	0.009 (0.015)	0.011 (0.013)	0.010 (0.011)	0.011 (0.011)	0.012 (0.011)	0.013 (0.011)
Firm age	−0.044*** (0.009)	−0.026*** (0.006)	−0.050*** (0.009)	−0.030*** (0.006)	−0.050*** (0.009)	−0.030*** (0.006)
Foreign ownership	0.002 (0.008)	0.006 (0.006)	−0.004 (0.008)	0.002 (0.007)	−0.004 (0.008)	0.002 (0.007)
SMEs	−0.003 (0.011)	−0.006 (0.009)	0.014 (0.011)	0.006 (0.008)	0.016 (0.011)	0.007 (0.008)
Small-sized city	−0.018 (0.017)	−0.022* (0.011)	−0.030 (0.019)	−0.031** (0.012)	−0.033* (0.018)	−0.032*** (0.012)
Medium-sized city	−0.017 (0.017)	−0.022* (0.012)	−0.021 (0.018)	−0.024* (0.013)	−0.022 (0.017)	−0.024* (0.013)
Asia	−0.004 (0.032)	−0.008 (0.024)	−0.002 (0.032)	−0.006 (0.024)	−0.002 (0.032)	−0.006 (0.025)
North Africa	0.065* (0.034)	0.046** (0.018)	0.085*** (0.031)	0.060*** (0.017)	0.087*** (0.030)	0.061*** (0.016)
Manufacturing firms	0.007 (0.009)	−0.062*** (0.012)	0.010 (0.010)	−0.059*** (0.012)	0.011 (0.009)	−0.059*** (0.012)
Observations	8941	8941	8941	8941	8941	8941
KP/Cragg-Donald			73.872	73.872	216.517	216.517
Root MSE	0.247	0.189	0.250	0.191	0.251	0.191

Notes: GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index. All models are estimated using a 2SLS estimator with robust standard errors clustered at the country level. Variables with zero or negative values are IHS transformed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, KP/Cragg–Donald and Kleibergen–Paap statistics test for weak instrument identification.

4. Results

This section reports the core findings of the analysis. In general, the empirical results reveal a significant effect of GI on financial performance, but a counterintuitive moderating effect of education, supporting the idea of stratified absorptive capacity and challenging conventional assumptions about the optimization of human capital. The section also explores how this dynamic varies across sectors, regions, and firm sizes, highlighting the contextual nature of the GI–education–performance relationship.

4.1. Main results

Table 4 reports the main results. For comparison, we also report the OLS estimation results for the model with an interaction term between GI and the education index.

First, we conducted thorough inspection tests to validate our model specification and estimation strategy. Across our models, variance inflation factors are below 2, suggesting no collinearity among variables. Likewise, the Breusch-Pagan test strongly rejected homoskedasticity ($p < 0.001$), leading us to employ robust and clustered standard errors. For our baseline 2SLS model, instrument strength diagnostics are consistent across both the sales and profit growth specifications. Shea's partial values indicate moderate relevance for GI (0.070) and GI × education index (0.091). The minimum eigenvalue statistic of 216.517 exceeds the critical threshold of 7.03 set by Stock and Yogo (2005) for two endogenous regressors and two excluded instruments, indicating that the instruments are sufficiently strong and that 2SLS estimates are robust to weak-identification concerns. Consistent with these findings,

Durbin-Wu-Hausman endogeneity tests offer more evidence in rejection of the null hypothesis of exogeneity for both profit growth ($\chi^2(2) = 12.534$, $p = 0.002$; $F(2, 8923) = 6.263$, $p = 0.002$) and sales growth ($\chi^2(2) = 14.843$, $p = 0.001$; $F(2, 8923) = 7.419$, $p = 0.001$), which shows that OLS estimates become inconsistent and indicates the need for IV strategy. Similarly, these diagnostics demonstrate that across models, the instruments are strong and valid and support the presence of endogeneity, thereby ensuring the robustness of our 2SLS estimation strategy. We also tested for nonlinear effects of workforce education by including GI × Education index² in our specifications, indicating the lack of a strict inverted-U nonlinear moderating effect.

As shown in Table 4, without accounting for endogeneity, our preliminary OLS + interaction results show a modest positive intensity effect of GI on sales growth and a non-significant effect on profit growth. However, when we apply the 2SLS + interaction specification (baseline model) to address endogeneity, the estimates shift markedly: GI exhibits stronger, positive, and statistically significant effects on both financial performance outcomes. Moreover, the interaction between GI and employee education becomes marginally significant for sales growth. The visual plot on the left panel of Fig. 2 confirms this significance. At low levels of employee education (approximately 0–20%), the marginal effect on sales growth is slightly above the significance threshold ($p \leq 0.01$). However, this benefit declines progressively with increasing education, becoming negligible at higher levels. This negative interaction indicates that firms with less-educated workforces derive significantly greater financial benefits from GI initiatives. Going forward, results for firm age and other significant control variables similarly reveal larger effects in the 2SLS specification than in OLS. This shift suggests that the 2SLS approach not only strengthens the estimated direct effects of GI but

Table 5
Analysis by gross profit margins – 2SLS/LIML regressions.

	Total sample	KIBS firms	North Africa
Independent variables	GPM	GPM	GPM
GI	−0.014 (0.036)	0.214** (0.100)	0.065 (0.085)
GI × Education index	−0.026 (0.043)	−0.087 (0.058)	0.140*** (0.025)
Education index	0.025 (0.017)	0.037 (0.026)	0.088* (0.049)
Competition	−0.013 (0.013)	−0.003 (0.026)	−0.041 (0.029)
Industry concentration	−0.006* (0.003)	0.017*** (0.005)	0.003 (0.011)
Exporter	0.008 (0.027)	−0.077* (0.047)	0.139*** (0.018)
Firm legal status	0.013 (0.015)	−0.013 (0.023)	0.121*** (0.025)
Firm age	−0.050*** (0.009)	0.007 (0.015)	−0.005 (0.008)
Foreign ownership	0.003 (0.019)	0.014 (0.021)	0.108*** (0.011)
SMEs	0.011 (0.015)	−0.010 (0.032)	−0.054*** (0.020)
Small-sized city	−0.041* (0.023)	−0.046* (0.024)	−0.066*** (0.013)
Medium-sized city	−0.027* (0.014)	−0.007 (0.019)	−0.045** (0.020)
Asia	0.006 (0.019)	0.086*** (0.032)	
North Africa	0.040 (0.033)	0.123** (0.049)	
Manufacturing firms	−0.437*** (0.017)		−0.367*** (0.046)
Observations	8941	534	1606
KP/Cragg-Donald	216.517	4.996	13.105
Root MSE	0.230	0.163	0.249

Notes: The specification for KIBS firms was estimated using the LIML estimator, and instrument strength was evaluated using the Cragg–Donald minimum eigenvalue statistic. The statistic (≈ 5) exceeds the Stock–Yogo critical values for 15% and 20% maximal size distortions (4.58 and 3.95, respectively), but falls below the 10% threshold (7.03), indicating borderline instrument strength. The LIML estimates should therefore be interpreted with caution. GI and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index. The LIML and 2SLS are estimated with robust standard errors clustered at the country level. Variables with zero or negative values are IHS transformed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, KP/Cragg–Donald and Kleibergen–Paap statistics test for weak instrument identification.

also reveals a meaningful moderating role of employee education and other controls. The 2SLS result reinforces Wooldridge (2010), who notes that ignoring endogeneity can bias estimates downward, underscoring

the importance of using appropriate econometric techniques.

4.2. Robustness checks

4.2.1. Testing mechanism robustness

To understand the mechanisms by which GI affects financial performance, we examine GPM as a separate outcome. We explored robustness across multiple contextual specifications. Statistically significant effects emerge in only three cases, as reported in Table 5, and we therefore focus our discussion on these results. Table 5 shows that GI's effect on margins is highly context-dependent. In the full sample, GI is non-significant for GPM ($\beta = -0.014$, n.s.), but exerts a direct, positive effect on GPM in Knowledge Intensive Business Service (KIBS) firms ($\beta = 0.214$, $p < 0.00$). Workforce education did not moderate these effects in the full sample, and the standalone education effect is non-significant ($\beta = 0.002$, n.s.). In North African markets, the mechanism differs fundamentally. GI alone does not significantly improve margins ($\beta = 0.065$, n.s.), but workforce education strongly and positively moderates the relationship ($\beta = 0.140$, $p < 0.01$). Fig. 3 shows a contrasting line: at low levels of workforce education (0–20%), the marginal effect of GI on GPM is small (approximately 0.07–1.46) and statistically insignificant. In contrast, at higher education levels (80–100%), GI has a strong and

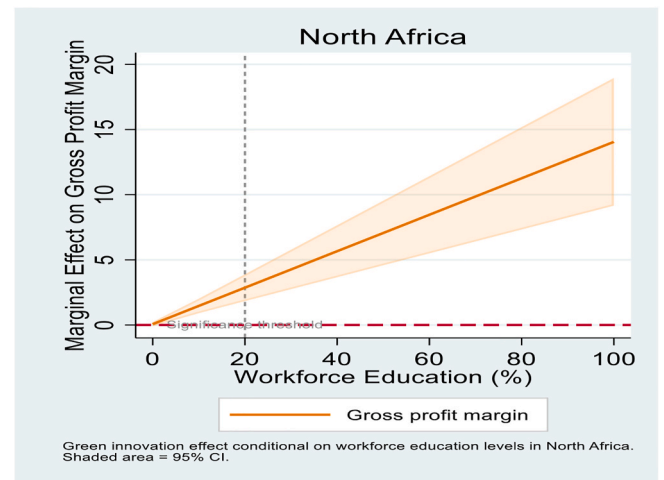


Fig. 3. Interaction effect of employee education levels on the GI-Gross profit margins relationship.

Note: X-axis: Workforce education index (0–1.5 scale; see Table 2 for details). Y-axis: Marginal effect of GI on sales and profit growth. Dashed line: significance threshold ($p \leq 0.01$). Shaded areas: 95% CIs. Fig. 3 applies only to the GPM specification in North Africa.

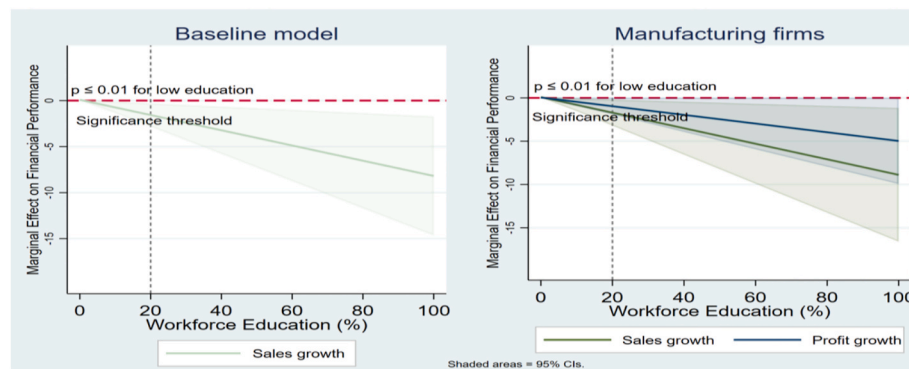


Fig. 2. Interaction effect of employee education levels on GI-sales/profit growth relationships.

Note: X-axis: Employee education index (0–1.5 scale; see Table 2 for details). Y-axis: Marginal effect of GI on sales and profit growth. Dashed line: significance threshold ($p \leq 0.01$). Shaded areas: 95% CIs. Fig. 2 applies to all models with significant interactions.

positive impact on GPM, reaching approximately +14.06. This indicates that a firm's absorptive capabilities determine whether it can leverage green initiatives to gain a premium advantage. The direct education effect ($\beta = 0.088$, $p < 0.10$) is only marginally significant, confirming that education's value lies primarily in its interaction with GI. This moderation pattern reveals fundamentally different value-creation mechanisms. In the full sample, GI did not improve margins, and education is insufficient to create cost value. In North African markets, GI alone is insufficient, whereas through education, higher human capital amplifies its profit value by building absorptive capacity.

4.2.2. Addressing self-selection bias

Beyond the endogeneity addressed by our IV approach, we test for potential self-selection bias, specifically, whether firms adopt GI based on unobserved characteristics associated with financial performance. Drawing on innovation management theory, we posit that managerial experience enhances a firm's ability to recognize and implement innovative opportunities, particularly in optimizing green practices (Hanson et al., 2016; Hatakeda et al., 2012; Teece, 2016). Regional innovation ecosystems also shape the selection and effectiveness of these practices by providing institutional support and contextual resources (Asheim and Gertler, 2005). Therefore, we posit that managerial experience, measured relative to the regional mean, influences both the likelihood of GI adoption (green innovators) and subsequent firm financial performance outcomes.

To assess potential sample selection bias, we implement a Heckman-type control-function correction combined with a 2SLS framework to address endogeneity in the GI measures. Government spending on education serves as an exclusion restriction, consistent with prior evidence linking public investment in education to innovation capacity (Castellacci and Natera, 2013; Leogrande and Costantiello, 2023). The baseline 2SLS estimates reported in Table 6 indicate that GI significantly improves both financial performance outcomes. The probit selection model is supported by the data ($\chi^2 = 5.08$, $p = 0.024$), and the first-stage F-statistics exceed the conventional threshold of 10. Following selection correction, the estimated coefficients on GI increase modestly for both performance measures. At the same time, the Inverse Mills Ratio (IMR) is statistically significant only in the profit growth model ($\beta = 0.066$, $p < 0.05$). Bootstrap standard errors based on 200 replications and clustered at the country level confirm the robustness of these findings: GI remains statistically significant across both financial performance outcomes, whereas the IMR is insignificant for sales growth ($\beta = 0.041$, $p > 0.10$) and only marginally significant for profit growth ($\beta = 0.066$, $p < 0.10$). Overall, the results suggest that GI has a positive, economically meaningful effect on firm financial performance, while evidence of selection bias is limited and statistically weak. Given the stability of the estimates and the absence of substantial selection effects across specifications, we rely on the baseline 2SLS estimates in the main analysis.

Having addressed unobserved selection through a 2SLS estimation, we conducted another robustness check using propensity score matching (PSM) to account for observable selection bias, following the same approach used to identify green innovators in the selection model. Fig. 4 illustrates the density distributions of propensity scores for treated (green innovators) and control firms (non-green innovators) across both financial performance outcomes—sales and profit growth. The distributions are closely centered around 0.6 and span from approximately 0.1 to 0.95, with near-undistinguishable shapes for both outcomes. This high degree of overlap confirms strong common support and suggests that the matching process achieves satisfactory covariate balance across specifications.

As shown in Table 7, 2SLS estimates from the baseline specification reveal a significant positive effect of GI on sales and profit growth. The PSM estimates also show statistically significant positive effects, though smaller in magnitude. These results suggest that observable characteristics explain part of the performance gains. However, the larger 2SLS coefficients imply that unobserved factors, such as managerial

Table 6

Regression results for the baseline specification – 2SLS and selection correction models.

Independent Variables	Sales Growth		Profit Growth	
	Baseline IV	Selection-Corrected	Baseline IV	Selection-Corrected
GI	0.106*** (0.038)	0.111*** (0.033)	0.052* (0.027)	0.069** (0.027)
Education index	0.002 (0.014)	−0.007 (0.013)	0.009 (0.012)	−0.005 (0.011)
GI × Education index	−0.066* (0.039)	−0.052* (0.031)	−0.018 (0.032)	−0.029 (0.031)
Inverse Mills Ratio		0.041 (0.032)		0.066** (0.031)
Bootstrap Results				
GI		0.111*** (0.033)		0.069** (0.027)
GI × Education index		−0.052 (0.034)		−0.029 (0.033)
Inverse Mills Ratio		0.041 (0.039)		0.069* (0.036)
Other controls	Included	Included	Included	Included
Observations	4875	8941	4875	8941
First-stage KP F-stat.	116.75	216.525	116.75	216.525

Notes: All models are estimated using a 2SLS estimator with robust standard errors clustered at the country level. GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index. The selection-corrected models include the Inverse Mills Ratio from the first-stage selection equation. Here, managerial experience (relative to the regional mean) captures innovation-oriented managerial capability and enters as a determinant of GI adoption. The lower panel reports bootstrap standard errors based on 200 resampling replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

capability or strategic orientation, may be driving additional benefits. Beyond the main effects, the interaction with workforce education (GI × Education index) shows marginally significant negative effects for sales growth under both methods, indicating that GI's performance benefits may be attenuated in firms with more highly educated workforces (Same interpretation as Fig. 2). The consistency in sign and direction across both financial performance outcomes and estimation methods across contexts complements the robustness of our findings. However, we only report estimates from the baseline specification for parsimony.

4.3. Heterogeneity assessments

4.3.1. Industry analysis: manufacturing vs. service firms

Table 8 presents the 2SLS estimates examining sectoral heterogeneity in the effects of GI on firm performance. GI significantly enhances sales and profit growth across manufacturing and service firms, with larger effects observed in the service sector. The education index shows no direct impact on financial performance outcomes. However, its interaction with GI reveals a notable divergence: in manufacturing, higher education levels significantly dampen GI's positive effects, whereas in services, no such moderation is evident. The right panel of Fig. 2 confirms this pattern, showing that firms with lower education index values—those with less educated workforces—experience the most considerable financial gains from GI initiatives. (Fig. 2 applies similarly to subsequent models). This differential moderating impact suggests sector-specific dynamics in how human capital interacts with innovation processes. Also, among control variables, firm age consistently predicts lower financial performance outcomes, consistent with organizational aging effects. Geographic location also matters: firms in

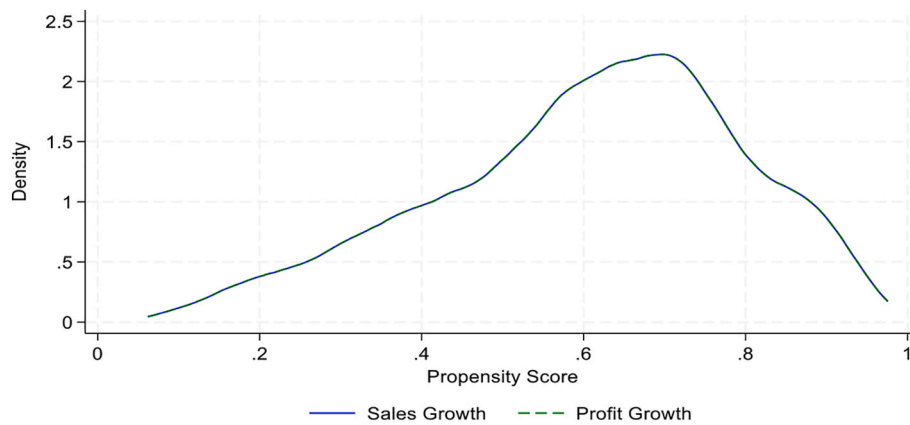


Fig. 4. Post-matching propensity score distributions for treated and control firms.

Notes: The x-axis reports the propensity score, defined as the estimated probability of treatment conditional on observed covariates used in the nearest-neighbor matching procedure. The y-axis shows the kernel density of propensity scores for matched treated and control firms, illustrating common support after one-to-one Propensity Score Matching.

Table 7

Comparison of 2SLS and Propensity – Score Matching Estimates for Baseline Specification (N = 8941 firms).

Method/Model	Coefficient	Std. Error	t-statistic	p-value
IV- 2SLS Estimates for Sales Growth				
GI	0.111	0.033	3.34	0.001
Education index	0.002	0.031	0.19	0.846
GI × Education index	−0.053	0.031	−1.69	0.091
IV- 2SLS Estimates for Profit Growth				
GI	0.070	0.027	2.57	0.010
Education index	0.011	0.010	0.98	0.327
GI × Education index	−0.030	0.031	−0.98	0.328
PSM Estimates for Sales Growth				
GI	0.030	0.008	3.88	0.001
Education index	0.015	0.010	1.50	0.134
GI × Education index	−0.019	0.012	−1.65	0.098
PSM Estimates for Profit Growth				
GI	0.015	0.006	2.54	0.011
Education index	0.018	0.008	2.32	0.020
GI × Education index	−0.013	0.009	−1.37	0.171

Notes: All models include the full set of controls and are estimated using a 2SLS estimator with robust standard errors clustered at the country level. GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index (*rd_std* × Education index). The upper panel reports 2SLS estimates; the lower panel reports PSM post-matching estimates with bootstrapped standard errors. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

North Africa outperform those in other regions, while those in smaller cities lag behind their counterparts in larger cities. Other firm-level characteristics, including competition, exporter status, ownership structure, and firm size, do not exhibit significant effects conditional on GI.

4.3.2. Sub-sectoral analysis

A structural analysis examining manufacturing (high-tech vs. low-tech) and service firms (traditional services vs. KIBS) is further conducted to test whether aggregate sectoral patterns hold across industry substructures, as presented in Table 9. The 2SLS regression results reveal substantial heterogeneity in the relationship between GI and firm performance across these four industry categories. In manufacturing sectors, GI is positively associated with both sales and profit growth, but is statistically stronger and more consistent in low-tech manufacturing. Meanwhile, in high-tech manufacturing, the association with sales

Table 8

Heterogeneity analysis by firm sectors – 2SLS regression.

Independent variables	Manufacturing firms		Service firms	
	Sales growth	Profit growth	Sales growth	Profit growth
GI	0.102*** (0.035)	0.047** (0.021)	0.123** (0.048)	0.109** (0.046)
Education index	0.011 (0.015)	−0.002 (0.011)	0.016 (0.017)	0.024 (0.016)
GI × Education index	−0.076** (0.036)	−0.047* (0.028)	−0.020 (0.055)	−0.005 (0.053)
Competition	0.007 (0.011)	0.001 (0.008)	0.001 (0.012)	−0.000 (0.011)
Industry concentration	−0.002 (0.003)	−0.002 (0.002)	0.002 (0.003)	0.002 (0.003)
Exporter	0.008 (0.013)	0.006 (0.010)	0.014 (0.020)	0.012 (0.018)
Firm legal status	0.009 (0.014)	0.008 (0.008)	0.014 (0.015)	0.013 (0.014)
Firm age	−0.049*** (0.009)	−0.020*** (0.004)	−0.051*** (0.010)	−0.046*** (0.009)
Foreign ownership	−0.004 (0.010)	0.004 (0.008)	−0.005 (0.017)	−0.003 (0.016)
SMEs	0.017 (0.012)	0.006 (0.007)	0.012 (0.019)	0.010 (0.018)
Small-sized city	−0.021 (0.020)	−0.023** (0.009)	−0.040* (0.020)	−0.035* (0.019)
Medium-sized city	−0.007 (0.020)	−0.014 (0.011)	−0.036** (0.018)	−0.033** (0.017)
Asia	−0.020 (0.029)	−0.006 (0.015)	−0.002 (0.042)	0.001 (0.039)
North Africa	0.068** (0.029)	0.033*** (0.010)	0.124*** (0.038)	0.120*** (0.035)
High-tech firms	0.014 (0.011)	0.007 (0.006)		
KIBS			0.012 (0.013)	−0.001 (0.011)
Observations	5155	5155	3786	3786
Cragg-Donald	145.982	145.982	70.458	70.458
Root MSE	0.244	0.143	0.261	0.245

Notes: GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index. All models are estimated using a 2SLS estimator with robust standard errors clustered at the country level. Variables with zero or negative values are IHS transformed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, KP/Cragg–Donald and Kleibergen–Paap statistics test for weak instrument identification.

Table 9
Heterogeneity analysis by industries – 2SLS regression.

Independent variables	High-Tech. Manufacturing		Low-Tech. Manufacturing		Traditional service firms		KIBS firms	
	Sales growth	Profit growth	Sales growth	Profit growth	Sales growth	Profit growth	Sales growth	Profit growth
GI	0.069* (0.041)	0.039 (0.028)	0.127*** (0.043)	0.047* (0.028)	0.118* (0.058)	0.095* (0.055)	0.290 (0.209)	0.300 (0.139)
Education index	−0.017 (0.019)	−0.001 (0.013)	−0.002 (0.018)	0.001 (0.012)	0.017 (0.022)	−0.019 (0.019)	−0.022 (0.019)	−0.016 (0.019)
GI × Education index	−0.043 (0.039)	−0.050 (0.032)	−0.116* (0.066)	−0.035 (0.043)	−0.030 (0.078)	−0.017 (0.074)	−0.093 (0.127)	−0.096 (0.116)
Competition	0.000 (0.015)	−0.010 (0.013)	0.014 (0.011)	0.014 (0.008)	−0.002 (0.012)	−0.004 (0.011)	0.051 (0.050)	0.048 (0.048)
Industry concentration	−0.010*** (0.003)	−0.006*** (0.002)	0.004 (0.005)	0.001 (0.002)	0.002 (0.003)	0.002 (0.003)	−0.001 (0.009)	0.003 (0.008)
Exporter	0.030* (0.017)	0.016* (0.009)	−0.017 (0.021)	−0.006 (0.018)	0.014 (0.018)	0.013 (0.015)	−0.055 (0.104)	−0.066 (0.099)
Firm legal status	0.009 (0.021)	0.014 (0.009)	0.005 (0.013)	0.001 (0.011)	0.015 (0.018)	0.015 (0.015)	−0.044 (0.036)	−0.035 (0.035)
Firm age	−0.046*** (0.010)	−0.019*** (0.005)	−0.049*** (0.010)	−0.020*** (0.010)	−0.047*** (0.010)	−0.043*** (0.010)	−0.072*** (0.020)	−0.059*** (0.018)
Foreign ownership	0.013 (0.019)	0.008 (0.012)	−0.025 (0.016)	−0.001 (0.011)	0.011 (0.010)	0.009 (0.010)	−0.072* (0.037)	−0.055 (0.034)
SMEs	0.008 (0.017)	0.001 (0.011)	0.023 (0.016)	0.010 (0.009)	0.007 (0.015)	0.005 (0.013)	0.052 (0.048)	0.042 (0.042)
Small-sized city	−0.012 (0.024)	−0.018* (0.010)	−0.026 (0.021)	−0.026** (0.011)	−0.044* (0.022)	−0.034* (0.018)	−0.039 (0.035)	−0.039 (0.035)
Medium-sized city	0.000 (0.024)	−0.013 (0.012)	−0.013 (0.020)	−0.016 (0.012)	−0.040** (0.018)	−0.037** (0.017)	−0.015 (0.031)	−0.011 (0.031)
Asia	−0.004 (0.032)	−0.007 (0.016)	0.008 (0.030)	−0.006 (0.016)	0.009 (0.043)	0.066 (0.058)	0.066 (0.058)	−0.044 (0.044)
North Africa	0.049 (0.031)	0.019 (0.013)	0.086*** (0.028)	0.045*** (0.011)	0.119*** (0.038)	0.104*** (0.036)	0.149*** (0.068)	0.234*** (0.063)
Observations	2540	2540	2615	2615	3252	3252	534	534
Cragg-Donald	65.269	65.269	72.819	72.819	62.509	62.509	4.996	4.996
Root MSE	0.245	0.147	0.139	0.139	0.259	0.243	0.288	0.271

Notes: GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index. All models are estimated using a 2SLS estimator with robust standard errors clustered at the country level. Variables with zero or negative values are IHS transformed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, KP/Cragg–Donald and Kleibergen–Paap statistics test for weak instrument identification.

growth is only marginally significant, suggesting a weaker but directionally similar pattern. Among service firms, traditional service providers also exhibit positive associations with both performance metrics, though only marginally. In contrast, the coefficients for KIBS firms are positive but statistically inconclusive, indicating that performance returns to GI may be more diffuse or contingent on complementary intangible assets in knowledge-intensive contexts.

The interaction between GI and the education index shows a negative moderation effect in low-tech manufacturing, though it is only weakly significant for sales growth. With the same interpretation in Fig. 2, this suggests that the GI benefits decrease as workforce education increases, possibly due to saturation effects or alternative innovation pathways. Beyond this effect, financial performance declines with age across all sectors, buttressing the idea that newer firms are more growth-oriented. Location effects are also evident, with smaller and medium-sized cities generally underperforming against larger urban centers, especially in manufacturing and traditional services. Low-tech manufacturing and service firms perform exceptionally well in North Africa, suggesting substantial geographical advantages in these regions. Export status has a slightly positive effect in high-tech manufacturing, and foreign ownership has a weak negative effect in KIBS. Industry concentration reduces performance in high-tech manufacturing, suggesting that, when innovation is used as a strategic response, competitive intensity can facilitate innovation-led growth in these sectors.

4.3.3. Geographic analysis

The positive relationship between GI and profit growth, shown in Table 10, is significant only at the marginal level in North Africa. From this result, GI may enhance profitability more than market expansion, possibly through operational efficiencies or niche positioning. In Europe, GI significantly enhances both financial performance and

outcomes, indicating a more favorable institutional and market environment for translating sustainability efforts into business outcomes. In contrast, Asian firms show no significant relationship between GI and either performance metric, indicating that environmental innovation may not yet be fully rewarded in this region.

The moderating role of human capital is observed only in Europe, not in North Africa or Asia. In Europe, the interaction between GI and the education index is significantly negative for sales growth, suggesting that the sales benefits of GI are stronger in firms with less educated workforces (same interpretation as Fig. 2). This may reflect sectoral positioning, where firms with lower human capital operate in markets where environmental credentials offer clearer differentiation. Alternatively, GI in these contexts may rely more on practical application and operational efficiency than on advanced technical expertise. For other control variables, firm age negatively affects financial performance outcomes across all regions. Industry concentration supports performance in North Africa but has negligible effects in other regions. While exporter status shows positive associations with profit growth in North Africa, firm legal form has favorable impacts on financial performance in both North Africa and Asia. Foreign ownership has a marginally negative effect on sales growth in North Africa only. SMEs operating in European firms benefit from financial performance, but not in North African and Asian firms. City size matters in Asia, where firms in smaller and medium-sized cities underperform. Manufacturing performs less than service firms across all regions, notably in North Africa and Europe.

4.3.4. SMEs vs. large firm analysis

Table 11 presents the 2SLS regression results, comparing the impact of GI on the financial performance of SMEs and large firms. For SMEs, GI is positively associated with sales growth, while its effect on profit growth is not significant. This suggests that GI may primarily support

Table 10
Heterogeneity analysis by geographic regions – 2SLS regression.

Independent variables	North Africa		Europe		Asia	
	Sales growth	Profit growth	Sales growth	Profit growth	Sales growth	Profit growth
GI	0.132 (0.109)	0.148* (0.089)	0.122*** (0.045)	0.053** (0.023)	0.029 (0.060)	0.016 (0.056)
Education index	0.007 (0.062)	0.013 (0.069)	0.021 (0.013)	0.017 (0.011)	0.000 (0.025)	0.016 (0.023)
GI × Education index	−0.052 (0.101)	−0.029 (0.101)	−0.097*** (0.031)	−0.043 (0.029)	0.107 (0.102)	0.096 (0.103)
Competition	−0.025* (0.014)	−0.034*** (0.012)	0.003 (0.009)	−0.001 (0.007)	0.019 (0.019)	0.018 (0.018)
Industry concentration	0.010*** (0.003)	0.008*** (0.002)	−0.001 (0.003)	−0.002 (0.002)	−0.010 (0.006)	−0.005 (0.004)
Exporter	0.014 (0.019)	0.049*** (0.012)	0.012 (0.015)	0.002 (0.009)	−0.011 (0.017)	−0.013 (0.014)
Firm legal status	−0.008 (0.023)	0.025*** (0.009)	−0.004 (0.010)	−0.009 (0.008)	0.062*** (0.024)	0.060** (0.027)
Firm age	−0.059*** (0.016)	−0.036*** (0.005)	−0.039*** (0.008)	−0.023*** (0.005)	−0.060*** (0.017)	−0.036*** (0.014)
Foreign ownership	−0.029* (0.017)	0.002 (0.011)	0.005 (0.009)	0.007 (0.007)	0.016 (0.033)	−0.003 (0.025)
SMEs	−0.011* (0.006)	−0.018* (0.010)	0.032*** (0.012)	0.018** (0.008)	−0.002 (0.033)	−0.002 (0.026)
Small-sized city	0.031 (0.040)	−0.004 (0.017)	−0.012 (0.015)	−0.009 (0.009)	−0.112* (0.061)	−0.097** (0.038)
Medium-sized city	0.038 (0.057)	0.001 (0.042)	−0.003 (0.010)	−0.004 (0.009)	−0.079** (0.037)	−0.065** (0.031)
Manufacturing firms	−0.015*** (0.003)	−0.106*** (0.021)	0.013 (0.010)	−0.054*** (0.008)	0.016 (0.029)	−0.051* (0.029)
Observations	1606	1606	5504	5504	1831	1831
Cragg-Donald	13.105	13.105	109.382	109.382	48.353	48.353
Root MSE	0.234	0.189	0.237	0.176	0.295	0.230

Notes: GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with *rd_std* and its interaction with the Education index. All models are estimated using a 2SLS estimator with robust standard errors clustered at the country level. Variables with zero or negative values are IHS transformed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, KP/Cragg–Donald and Kleibergen–Paap statistics test for weak instrument identification.

market expansion in smaller firms, possibly through differentiation or niche positioning. Conversely, large firms benefit more generally: GI significantly enhances both financial performance outcomes for larger firms, with stronger coefficients and higher significance levels. Such results suggest that the scale advantages, brand leverage, or more integrated sustainability strategies enjoyed by larger firms make them more effective at converting GI into improved financial performance. Whereas the interaction term (GI × education index) is insignificant for SMEs, it has a negative, significant coefficient for large firms, suggesting that the sales benefits of GI dissipate in firms with more educated workforces. (Interpretation follows Fig. 2). This outcome may reflect more competitive or sophisticated markets where environmental features are embedded within broader innovation strategies rather than serving as standalone differentiators.

Control variables, such as firm age, are negatively associated with financial performance outcomes, with slightly stronger effects among SMEs. Exporter status is positively associated with performance in SMEs but shows no significant effect in large firms. Firm legal status and foreign ownership have limited influence across both groups. Regarding location effects, being situated in smaller cities is significantly associated with lower profit growth compared to metropolitan cities. Regional effects also vary: firms in North Africa show stronger, significant positive associations with financial performance outcomes than European firms across firm sizes, whereas Asian firms show no significant impact. Finally, manufacturing firms exhibit significantly lower profit growth in both SMEs and large firms, reinforcing the notion that service-oriented sectors may be better positioned to capitalize on GI.

5. Discussion and implications of the study

Section 5.1 (Discussion) opens this section and is subdivided into three parts that interpret the results: the GI-financial performance

linkage (5.1.1), an employee education “less is more” paradox (5.1.2), and other control variables (5.1.3). Section 5.2 presents the implications of the above findings for theory (5.2.1), management (5.2.2), and policymaking (5.2.3). A limitations discussion follows to round out this section, after which a brief final section draws on salient findings to propose future research directions.

5.1. Discussion

Previous studies affirm that green practices drive sustainable growth by enhancing competitiveness and meeting regulatory and consumer expectations (Shuwaikh et al., 2023; Banelienė and Strazdas, 2023; Acemoglu et al., 2023). We extend this evidence, showing that GI improves financial performance in this context. However, the magnitude of this impact appears to depend on employee education, a key human capital resource that shapes firms' ability to absorb and apply sustainability practices. We therefore examine how employee education moderates GI's financial effects across sectors, firm sizes, and geographic regions, offering new insights into the conditions under which GI may optimize financial returns.

5.1.1. Green innovation and financial performance

Our findings for Hypothesis 1 confirm that GI significantly improves financial performance across diverse organizational contexts, aligning with prior research like Shuwaikh et al. (2023). Also, this finding aligns with Andersén (2021) and Silvério et al. (2025) in conceptualizing GI as a dynamic capability within the RBV. Moreover, it aligns with Hundloe (2021) and Purvis et al. (2018) on a niche-level driver of systemic change in STT. The STT emphasizes that economies evolve through multilevel reconfigurations involving technology, institutions, and policy (Zhai and Chang, 2018). GI operates within this framework by enabling firms to respond to environmental pressures while achieving

Table 11
Heterogeneity analysis by firm size – 2SLS regression.

Independent variables	SMEs		Large firms	
	Sales growth	Profit growth	Sales growth	Profit growth
GI	0.078** (0.037)	0.039 (0.034)	0.164*** (0.052)	0.110*** (0.036)
Education index	0.005 (0.016)	0.013 (0.015)	0.035 (0.024)	0.026 (0.018)
GI × Education index	−0.018 (0.045)	−0.011 (0.044)	−0.126* (0.068)	−0.065 (0.054)
Competition	0.007 (0.011)	−0.001 (0.008)	−0.013 (0.012)	−0.005 (0.011)
Industry concentration	0.001 (0.003)	0.001 (0.002)	−0.007 (0.006)	−0.006 (0.005)
Exporter	0.025* (0.015)	0.021* (0.011)	−0.012 (0.016)	−0.012 (0.011)
Firm legal status	0.000 (0.008)	0.002 (0.007)	0.040 (0.026)	0.042 (0.027)
Firm age	−0.053*** (0.010)	−0.031*** (0.007)	−0.042*** (0.009)	−0.028*** (0.007)
Foreign ownership	0.002 (0.013)	0.016 (0.010)	−0.001 (0.012)	−0.005 (0.010)
Small-sized city	−0.029 (0.019)	−0.027** (0.012)	−0.027 (0.019)	−0.031** (0.015)
Medium-sized city	−0.023 (0.021)	−0.027* (0.014)	−0.005 (0.016)	−0.009 (0.013)
Asia	−0.019 (0.038)	−0.018 (0.028)	0.037 (0.023)	0.020 (0.019)
North Africa	0.079** (0.033)	0.050*** (0.016)	0.100*** (0.027)	0.085*** (0.020)
Manufacturing firms	0.016* (0.008)	−0.055*** (0.016)	0.003 (0.003)	−0.062*** (0.020)
Observations	6213	6213	2728	2728
Cragg-Donald	118.668	118.668	79.656	79.656
Root MSE	0.258	0.196	0.230	0.176

Notes: GI (green innovation) and the interaction term (Education index × GI) are treated as endogenous and instrumented with rd_std and its interaction with the Education index. All models are estimated using a 2SLS estimator with robust standard errors clustered at the country level. Variables with zero or negative values are IHS transformed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, KP/Cragg–Donald and Kleibergen–Paap statistics test for weak instrument identification.

economic goals. Our results reflect this dynamic: firms adopting GI benefit from regulatory incentives, improved reputation, and access to sustainability-conscious markets (Holzner and Wagner, 2022; Acemoglu et al., 2023). From the RBV perspective, GI enables a firm to reorganize internal resources, such as technologies, processes, and knowledge, into valuable and inimitable capabilities that contribute to environmental and financial performance (Andersén, 2021; Silvério et al., 2025). In this context, energy efficiency, emissions monitoring, and waste reduction are GI practices that reduce operational costs, demonstrate an environmental commitment, enhance stakeholder trust, and differentiate a firm from competitors (Jia et al., 2021).

Sectoral analysis reveals stronger GI effects in service firms than in manufacturing. Service incorporates environmental innovation as intangible assets such as brand equity and customer involvement (Colombelli et al., 2015). Manufacturing firms incur higher implementation costs and longer paybacks for innovations, particularly in high-tech firms. Low-tech manufacturing firms, on the other hand, show a stronger association with performance, suggesting that sustainability offers a clearer differentiator in markets where it remains a new feature. Traditional service providers show positive associations between GI and both sales and profit growth, though these are only marginally significant. This suggests that GI contributes to performance even in less innovation-intensive service contexts, likely through operational improvements and customer responses to sustainability signals. KIBS, on

the other hand, exhibit positive but statistically inconclusive coefficients. This pattern implies that GI's financial returns in KIBS may depend on complementary intangible assets—such as advanced human capital, client customization, or absorptive capacity—rather than GI alone. These findings align with the RBV and STT, which emphasize that innovation outcomes are shaped by the interaction between firm capabilities and institutional environments (Hundloe, 2021; Frenken et al., 2023; Silvério et al., 2025).

That being said, geographic heterogeneity underlies explaining institutional quality. The European firms gain significantly from GI, consistent with mature regulatory frameworks and innovation ecologies that reward sustainability (Crescenzi and Rodríguez-Pose, 2011; Frenken et al., 2023). On the contrary, significant GI effects are absent for Asian firms; this is clearly due to underdeveloped institutional environments and weaker market incentives (Bate et al., 2023). For North Africa, modest gains in profitability indicate that GI improves operational efficiency rather than market expansion in developing economies (Gebrue, 2024). Firm size moderates the association of GI with performance. At the aggregate level, large firms tend to benefit more from GI, buttressing Knott and Vieregger's (2015) argument that they leverage scale economies and integrated sustainability strategies to drive performance. SMEs exhibit positive sales effects with limited profit gains, confirming Sen et al. (2023), who posit that smaller firms leverage their agility to turn limited resources and absorptive capacity into performance benefits. These findings support Andersen's (2021) perspective that GI is essentially a strategic capability whose financial returns depend on the organizational structure and its alignment with the operational context. Overall, the contingent business case for our findings is that GI builds market position more consistently than it improves profitability. Revenue effects appear broadly, while profit gains concentrate in contexts with scale advantages (large firms), institutional support (Europe), or lower implementation barriers (services). This asymmetry—market validation without uniform profit payoff—reflects offsetting mechanisms that aggregate profitability ratios would obscure.

5.1.2. Employee education and the “less is more” paradox

Hypothesis 2 is supported by a counterintuitive, theoretically grounded finding: firms with lower levels of employee education appear to derive greater financial benefits from GI than those with highly educated workforces. This pattern supports our view of the ‘less is more’ hypothesis, traceable to ACT and cognitive load perspectives (Leal-Rodríguez et al., 2013; Leppink and Van den Heuvel, 2015). From the ACT perspective, Cohen and Levinthal, (1990) and Zahra and George (2002) contend that, while highly educated workers may become cognitively overloaded or locked into rigid knowledge structures, the benefits of this education decrease. This perception, backed by Bunjak et al. (2021), Stajkovic and Stajkovic (2025), and Yuwono (2021), limits creative problem-solving and cross-functional collaboration. Our findings, in line with Aghion et al. (2009), indicate that employees with intermediate-level qualifications tend to have practical experience and adaptability that improve operational learning. These are practice capabilities that support firm-switching events toward sustainability, involving multilevel transformations of policy, markets, and culture (Hundloe, 2021). This paradox suggests that valuable resources need not be highly formalized; tacit and flexible capabilities may be strategic to capture the economics of GI from an RBV perspective.

Sectoral and industry-level analyses continue to exhibit the education-GI interaction pattern: lower levels of employee education appear to drive greater financial benefits from GI than highly educated workforces in manufacturing, but not in services. This result is plausible because, in manufacturing, especially in low-tech segments, GI often targets tangible process improvements such as energy efficiency and waste management. These improvements depend more on tacit knowledge and experiential learning than on formal educational qualifications. This view aligns with Human Capital Theory (Becker, 1975) and absorptive capacity perspectives (Zahra and George, 2002), pointing out

that experiential learning is fundamental to knowledge assimilation and application. Within this context, the current findings support the notion advanced by Porter and Linde (1995) that capital-intensive innovation and automation can reduce the marginal contribution of human capital at the operational level. It also buttresses Acemoglu et al.'s (2006) argument that high fixed costs, together with slow diffusion rates (in our context, GI), restrain the short-term returns to education. As high-tech manufacturing and KIBS do not show this pattern, they benefit from advanced education, given that radical, knowledge-intensive innovations demand specialized expertise (Acemoglu et al., 2006; Miozzo and Soete, 2001). This divergence shows that education–innovation outcomes depend on industry characteristics and innovation types. Hence, low-tech firms can achieve comparable GI financial performance through experiential learning, consistent with Horbach's (2008) view of firm-level routines and tacit knowledge. In a way, this finding aligns with Demirel and Kesidou (2011), showing that targeted policies and organizational practices can substitute for formal education in building GI capabilities.

Geographic variation further supports the institutional mediation argument. In North Africa and Asia, the moderating effect of employee education is insignificant for the relationships between GI and sales and profit growth, likely reflecting the institutional challenges documented in these regions (Gebrue, 2024). In contexts of institutional voids, human capital must serve multiple compensatory roles (Khanna and Palepu, 2010), potentially diluting its moderating effect on specific innovation–performance relationships. As such, when examining GPM as an outcome, a different pattern emerges. While workforce education strongly and positively moderates GI's effect on GPM. This divergence suggests different mechanisms for value capture. Educated employees may enable firms to convert green initiatives into higher margins through premium-pricing know-how or operational efficiency. Even then, these margin advantages do not necessarily translate into faster growth rates. This is consistent with North African firms operating in resource-constrained environments where absorptive capacity (education) enables value extraction rather than market expansion. In contrast, the significant education–GI interaction for growth in European markets reflects how external institutional support can substitute for internal human capital in driving scale.

Firm size analysis reveals that the education paradox is confined to large firms. In these organizations, multiple layers and specialized departments can create cognitive silos when overpopulated with highly credentialed staff (Leal-Rodríguez et al., 2013). SMEs, by nature of their flatter structures and the multifunctional nature of their roles, integrate knowledge naturally and avoid fragmentation (Sen et al., 2023). Employees in SMEs, even highly qualified ones, stay closer to operational realities and, thus, maintain contextual awareness and adaptability. These findings help resolve the tension between human capital theory and cognitive load theory. Education improves absorptive capacity but also introduces processing constraints and coordination liabilities when excessively concentrated. In this regard, firms must trade off the benefits of formal education against the risks of excessive thinking and compartmentalized knowledge. According to Shafiullah et al. (2021), this perception becomes even more important in sustainability-related changes, where the effects of new ideas depend on the extent to which organizational skills align with institutional and market conditions.

5.1.3. Other controls

The observed effects of other control variables align with several established frameworks. Firm age negatively impacted financial performance, and supports Akben-Selcuk (2016), who argued that older firms may encounter structural inertia and reduced adaptability. With a marginally positive effect, exporter status is consistent with endogenous growth theory, which posits that internationalization is linked to productivity gains and innovation spillovers (Antonelli, 2019). Findings on geographic location effects underpin regional innovation systems theory, which suggests that innovation incidence reflects the density of

institutions and the knowledge infrastructure within regions (Asheim and Gertler, 2005; Crescenzi and Rodríguez-Pose, 2011). A firm's legal form affects profitability through facilitating enhanced access to capital and strategic flexibility, as described in corporate governance theory (Stowers, 2024). In line with Eide et al. (2021), negative associations between foreign ownership and financial performance, especially within the confines of KIBS in North Africa, suggest that foreign investors may prioritize global efficiency over local adaptation, potentially hindering a firm's ability to tailor GI to regional contexts. These findings underscore how ownership structure and market dynamics condition GI's financial returns across sectors and regions. Industry concentration has mixed effects: it appears to improve performance in North Africa, likely because dominant companies leverage scale and institutional advantages. Still, it dampens outcomes in high-tech manufacturing, illustrating Schumpeterian competition theory, which posits that while moderate competition is conducive to innovation, excessive market concentration may decrease dynamic efficiency and dampen innovation incentives (Aghion et al., 2005).

5.2. Implications of the study

5.2.1. Theoretical implications

This study offers three interlinked theoretical contributions that extend our understanding of how human capital interacts with GI. First, it refines the RBV by showing that the optimal value of GI is not intrinsic but contingent on complementary human capital configurations. Contrary to traditional RBV logic, firms with high concentrations of highly educated employees may experience reduced financial returns from GI due to cognitive overload and siloed thinking. This introduces resource composition complexity as a boundary condition to Barney's (1991) framework (Andersén, 2021; Silvério et al., 2025). That is, competitive advantage stems from strategically balanced educational diversity that fosters knowledge flow and resource complementarity (Domadenik et al., 2020; Wales et al., 2013). Second, the study extends ACT (Zahra and George, 2002) by documenting a negative moderation of employee education on the relationship between GI and financial performance outcomes. While education enhances potential absorptive capacity, excessive formal credentials can constrain realized absorptive capacity, limiting firms' ability to transform and exploit knowledge (Leppink and Van den Heuvel, 2015). This finding supports Aghion et al.'s (2009) perspective, leading to the concept of stratified absorptive capacity. This perception aligns with the view that optimal innovation performance depends on formal education, tacit knowledge, and experiential learning. Third, the current findings advance STT by revealing the multilevel boundary conditions under which GI drives systemic change (Hundloe, 2021). The financial impact of GI varies by sector, region, and firm size. GI enhances performance in services more than in manufacturing, is stronger in Europe than in Asia, and larger firms feel the impact more than SMEs. This result requires a contextualized transitions framework that embeds institutional quality, sectoral dynamics, and organizational architecture.

5.2.2. Managerial implications

Our findings reveal that GI's returns across financial performance outcomes are more akin to a market-access investment for maintaining or growing market share, rather than for immediate margin improvement. However, it necessitates differentiated strategic approaches. In resource-constrained contexts such as North Africa and parts of Asia, managers should prioritize leveraging workforce absorptive capacity to improve margins from GI, as growth effects may be more limited or slower to materialize. In low-tech manufacturing, where lower employee education levels generate superior GI returns on growth, managers should prioritize experiential learning and vocational training. They should empower frontline workers who possess tacit knowledge of production processes rather than pursuing credentials for their own sake. Service sector managers should integrate GI into

customer-facing operations. They should develop employees' abilities to communicate sustainability benefits and engage customers as partners in green value creation. Our results suggest services capture GI value through relational capital, such as brand equity and customer involvement. For KIBS, GI returns depend on complementary intangible assets rather than GI alone. Managers should invest in organizational capabilities—proprietary tools, knowledge management systems, and client customization processes—that enable highly educated professionals to apply their expertise across diverse contexts efficiently. High-tech manufacturing managers should maintain emphasis on advanced education for radical innovations. However, they must implement cross-functional collaboration mechanisms to prevent cognitive silos among highly credentialed staff. Large firms must be particularly attentive to knowledge fragmentation across specialized departments. In contrast, SMEs should leverage their natural agility and flatter structures to rapidly implement incremental GIs aligned with existing operational capabilities.

5.2.3. Policy implications

The findings of the current work suggest that policymakers should avoid a generalized policy framework. They must model differentiated support mechanisms that account for sectoral, regional, and firm-size differences. For manufacturing, notably low-tech segments, policies should support vocational training programs, apprenticeships, and worker participation initiatives. These firms should develop practical green skills rather than emphasizing academic credentials alone. Service sectors require support for training that integrates environmental awareness with customer engagement capabilities. Knowledge-intensive sectors benefit from funding for collaborative platforms, knowledge repositories, and technology transfer mechanisms. Regional policies must acknowledge institutional variation. In Europe, policymakers should continue strengthening regulatory frameworks and collaborative innovation networks. In Asia and North Africa, where institutional voids constrain GI adoption, policies should prioritize the development of internal firm capabilities. This includes technical assistance, industry associations, and targeted support for operational efficiency improvements that generate immediate financial benefits. Education policy should balance skill learning and formal knowledge, with adaptations according to different contexts. Besides formal academic qualifications, it should be modeled to support other routes of learning, such as vocational education and lifelong training. Size-specific policies will make accessible training and financial support available to resource-constrained SMEs. They should also encourage large firms to implement mechanisms for knowledge integration that bridge departmental silos. Fundamentally, both firm managers and policymakers must acknowledge that valuable GI capabilities need not be too formalized. They must view that tacit knowledge, experiential learning, and practical problem-solving skills are complementary strategic assets that vary across sectors and institutions.

5.3. Limitations of the study

Despite the important contributions of this study, several limitations remain. While documenting a positive association between GI and financial performance outcomes, future research might decompose how different types of GI interact with employee education to shape divergent performance outcomes. The reliance on cross-sectional data limits understanding of dynamic adjustment processes; it is assumed that longitudinal designs can uncover such adjustments. In our case, focusing on sales and profit growth comprises two very significant strategic dimensions. However, it does not allow us to assess net profitability or fully observe GI-related implementation and education costs, which may differ substantially across contexts. This is especially evident for North African markets, where the margin gains from GI seem contingent on education, suggesting a heterogeneous structure. Future studies using return ratios (ROA, ROE) or productivity (e.g., value added per person)

might help establish a link between the gains from GI and efficiency changes. Similarly, though education differentially moderates GI returns geographically, we cannot fully establish the reasons for this geographical specificity, even with IV methods to account for endogeneity. Future research combining richer cost data, institutional indicators, and detailed innovation characteristics could better isolate these causal pathways. Finally, limitations in measuring employee education and other employment categories, as well as the absence of sampling weights, may constrain generalizability. Our work, therefore, calls for broader human capital measures and for design-based inference in future work.

6. Conclusions

This study investigates how GI affects firm financial performance across diverse organizational and regional contexts, and whether employee education levels shape this relationship. Drawing on data from 8941 firms across 42 countries and employing a 2SLS estimation strategy, the study demonstrated that GI generally contributes positively to financial performance. However, this impact is far from uniform. Sectoral, regional, and organizational heterogeneity significantly condition the strength and direction of GI's financial performance. GI yields more substantial financial returns in European firms and in traditional manufacturing and service sectors. Meanwhile, its effects are either weaker or statistically insignificant in Asia, high-tech manufacturing, and KIBS firms. These key insights point to institutional maturity, sectoral structure, and resource configurations as important considerations, and echo theoretical insights from the RBV, ACT, and the literature on sustainability transitions.

A notable and somewhat counterintuitive finding is the “less is more” dynamic: firms with lower levels of employee education often realize the most favorable financial payoffs from GI. This outcome suggests that absorptive capacity, characterized by the distribution of practical expertise and flexible knowledge structures across educational tiers, outperforms more education-intensive configurations. This is especially true in contexts where operational flexibility and practical application are crucial. To this end, firm managers must not define GI strategies based on the assumption that all education, including higher education, is a panacea for improving GI performance, but rather based on matching employees' capabilities to the contextual realities of enterprises. Also, policymakers should invest in competency-based training systems built around layers of complementary skills within new institutional ecosystems. By implication, there is a need for future research exploring how various types of GI interact with different human capital profiles and institutional settings to shape financial trajectories. Similarly, it requires that panel studies with better measures of human capital, including tacit knowledge and cross-functional learning, can provide improved insights into how GI capability builds over time and contributes to continued economic growth across various contexts.

CRedit authorship contribution statement

Olugbenga Michael Adewumi: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.
Dengjun Zhang: Writing – review & editing, Validation, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2026.147857>.

Data availability

The data from the World Bank Enterprise Surveys that support our findings in this study are available at: <https://login.enterprisesurveys.org/content/sites/financeandprivatesector/en/signin.html>

The data used regarding development status analysis in our study is obtained from the World Economic Situation and Prospects 2023: Economic Analysis and Policy.

References

- Acemoglu, D., Aghion, P., Zilibotti, F., 2006. Distance to frontier, selection, and economic growth. *J. Eur. Econ. Assoc.* 4, 37–74. <https://doi.org/10.1162/jeea.2006.4.1.37>.
- Acemoglu, D., Aghion, P., Barrage, L., Hémous, D., 2023. Green innovation and the transition toward a clean economy. Peterson inst. Int. Econ. Working Pap. <https://doi.org/10.2139/ssrn.4816734>, 23–14.
- Aghion, P., Bloom, N., Blundell, R., Griffith, R., Howitt, P., 2005. Competition and innovation: an inverted-U relationship. *Q. J. Econ.* 120, 701–728.
- Aghion, P., Boustian, L., Hoxby, C., Vandenbussche, J., 2009. The causal impact of education on economic growth: evidence from US. Brookings Pap. Econ. Activ. 1, 1–73.
- Aghion, P., 2018. Innovation and growth from a Schumpeterian perspective. *Rev. Écon. Polit.* 128, 693–711. <https://doi.org/10.3917/redp.285.0693>.
- Aihounton, G.B., Henningsen, A., 2021. Units of measurement and the inverse hyperbolic sine transformation. *Econ. J.* 24 (2), 334–351. <https://doi.org/10.1093/ectj/utaa032>.
- Akben-Selcuk, E., 2016. Does firm age affect profitability? Evidence from Turkey. *Int. J. Econ. Sci.* 5, 1–14. <https://doi.org/10.20472/es.2016.5.3.001>.
- Aloqaily, A., 2023. The effects of green human resource on employees' green voice behaviors towards green innovation. *ABAC J* 43, 377–397. <https://doi.org/10.59865/abacj.2023.62>.
- Andersén, J., 2021. A relational natural-resource-based view on product innovation: the influence of green product innovation and green suppliers on differentiation advantage in small manufacturing firms. *Technovation* 104, 102254. <https://doi.org/10.1016/j.technovation.2021.102254>.
- Antonelli, C., 2019. The creative response and international trade. *Struct. Change Econ. Dynam.* 51, 445–452. <https://doi.org/10.1016/j.strueco.2019.03.002>.
- Asheim, B., Gertler, M.S., 2005. The geography of innovation: regional innovation systems. In: Fagerberg, J., Mowery, D.C., Nelson, R.R. (Eds.), *The Oxford Handbook of Innovation*. Oxford Univ. Press, Oxford, pp. 291–317.
- Banelienė, R., Strazdas, R., 2023. Green innovation for competitiveness: impact on GDP growth in the European Union. *Contemp. Econ.* 17, 92–108. <https://doi.org/10.5709/ce.1897-9254.501>.
- Barba-Aragón, M.I., Jiménez-Jiménez, D., 2024. Is training a green innovation driver? The mediating role of knowledge acquisition. *J. Knowl. Manag.* 28, 463–483. <https://doi.org/10.1108/jkm-10-2022-0818>.
- Barney, J., 1991. Firm resources and sustained competitive advantage. *J. Manag.* 17, 99–120. <https://doi.org/10.1177/014920639101700108>.
- Bate, A.F., Wachira, E.W., Danko, S., 2023. The determinants of innovation performance: an income-based cross-country comparative analysis using the global innovation index (GII). *J. Innov. Entrep.* 12, 1–19. <https://doi.org/10.1186/s13731-023-00283-2>.
- Becker, G.S., 1975. *Human Capital: a Theoretical and Empirical Analysis*. National Bureau Econ. Res., New York.
- Bentham, G., 2024. Sustainable technology: a competitive advantage for businesses. Capgemini Norway. <https://www.capgemini.com/no-no/insights/expert-perspectives/sustainable-technology-a-competitive-advantage-for-businesses>.
- Bogliacino, F., Pianta, M., 2013. Profits, R&D, and innovation—a model and a test. *Ind. Corp. Change* 22, 649–678. <https://doi.org/10.1093/icc/dts028>.
- Boschma, R., Gianelle, C., 2014. Regional branching and smart specialisation policy. JRC88242. Joint Research Centre, European Commission.
- Buhl, A., Blazejewski, S., Dittmer, F., 2016. The more, the merrier: why and how employee-driven eco-innovation enhances environmental and competitive advantage. *Sustainability* 8, 946. <https://doi.org/10.3390/su8090946>.
- Bunjak, A., Cerne, M., Popović, A., 2021. Absorbed in technology but digitally overloaded: interplay effects on gig workers' burnout and creativity. *Inf. Manag.* 58, 103533. <https://doi.org/10.1016/j.im.2021.103533>.
- Bronzini, R., Piselli, P., 2016. The impact of R&D subsidies on firm innovation. *Res. Pol.* 45, 442–457. <https://doi.org/10.1016/j.respol.2015.10.008>.
- Castellacci, F., Natera, J.M., 2013. The dynamics of national innovation systems: a panel cointegration analysis of the co-evolution between innovative capability and absorptive capacity. *Res. Pol.* 42, 579–594. <https://doi.org/10.1016/j.respol.2012.10.006>.
- Choi, Y.M., Park, K., 2019. Foreign ownership, agency costs, and long-term firm growth: evidence from Korea. *Sustainability* 11, 1599. <https://doi.org/10.3390/su11061599>.
- Clougherty, J.A., Duso, T., Muck, J., 2016. Correcting for self-selection-based endogeneity in management research. *Organ. Res. Methods* 19, 286–347. <https://doi.org/10.1177/1094428115619013>.
- Cohen, W.M., Levinthal, D.A., 1990. Absorptive capacity: a new perspective on learning and innovation. *Adm. Sci. Q.* 35, 128–152. <https://doi.org/10.2307/2393553>.
- Cohen, W.M., Klepper, S., 1992. The trade-off between firm size and diversity in the pursuit of technological progress. *Small Bus. Econ.* 4, 1–14. <https://doi.org/10.1007/bf00402211>.
- Colombelli, A., Krafft, J., Quatraro, F., 2015. Eco-innovation and firm growth: do green gazelles run faster? Microeconomic evidence from a sample of European firms. *WWWforEurope Working Pap.* 88.
- Consoli, D., Marin, G., Marzocchi, A., Vona, F., 2016. Do green jobs differ from non-green jobs in terms of skills and human capital? *Res. Pol.* 45, 1046–1060. <https://doi.org/10.2139/ssrn.2744521>.
- Crescenzi, R., Rodríguez-Pose, A., 2011. *Innovation and Regional Growth in the European Union*. Springer, Berlin.
- Crook, T.R., Todd, S.Y., Combs, J.G., Woehr, D.J., Ketchen Jr., D.J., 2011. Does human capital matter? A meta-analysis of the relationship between human capital and firm performance. *J. Appl. Psychol.* 96, 443–456. <https://doi.org/10.1037/a0022147>.
- Dangelico, R.M., 2016. Green product innovation: where we are and where we are going. *Bus. Strat. Environ.* 25, 560–576. <https://doi.org/10.1002/bse.1886>.
- Demirel, P., Kesidou, E., 2011. Stimulating different types of eco-innovation in the UK: government policies and firm motivations. *Ecol. Econ.* 70, 1546–1557. <https://doi.org/10.1016/j.ecolecon.2011.03.019>.
- Domadenik, P., Pastore, F., Koman, M., Redek, T., 2020. Innovation for a greener and more profitable future: a conceptual approach. In: Nesticó, A., Maselli, G. (Eds.), *Challenges on the Path Toward Sustainability in Europe*. Emerald, Bingley, pp. 127–145.
- Eide, L.S., Erraia, J., Grimsby, G., 2021. Industry concentration and profitability in Europe: the case of Norway. *Jahrbücher Natlökonomie Stat.* 241, 577–622. <https://doi.org/10.1515/jbnst-2020-0048>.
- European Commission, 2011. *Roadmap to a Resource Efficient Europe*. COM (2011) 0571. European Commission, Brussels.
- FasterCapital, 2024. Profit contribution: the role of profit contribution in business growth and sustainability. <https://fastercapital.com/content/Profit-Contribution-The-Role-of-Profit-Contribution-in-Business-Growth-and-Sustainability.html>.
- Frenken, K., Neffke, F., Van Dam, A., 2023. Capabilities, institutions and regional economic development: a proposed synthesis. *Camb. J. Reg. Econ. Soc.* 16, 405–416. <https://doi.org/10.1093/cjres/rsad021>.
- Gaio, C., Henriques, R., 2018. Are large firms more profitable than small and medium firms in the European Union? *Eur. J. Manag. Stud.* 23, 25–48.
- Gebrue, N., 2024. Institutional quality and economic growth in upper-middle-income African countries. *African J. Econ. Manag. Stud.* <https://doi.org/10.1108/ajems-01-2024-0044>.
- Grant, R.M., 1991. The resource-based theory of competitive advantage: implications for strategy formulation. *Calif. Manag. Rev.* 33, 114–135. <https://doi.org/10.2307/41166664>.
- Guinot, J., Barghouti, Z., Beltrán-Martín, I., Chiva, R., 2023. Corporate social responsibility toward employees and green innovation: exploring the link in the tourism sector. *Green Finance* 5, 298–320. <https://doi.org/10.3934/gf.2023012>.
- Hanson, D., Hitt, M.A., Ireland, R.D., Hoskisson, R.E., 2016. *Strategic Management: Competitiveness and Globalization*. Cengage, Australia.
- Hatakeda, T., Kokubu, K., Kajiwara, T., Nishitani, K., 2012. Factors influencing corporate environmental protection activities for greenhouse gas emission reductions: the relationship between environmental and financial performance. *Environ. Resour. Econ.* 53, 455–481. <https://doi.org/10.1007/s10640-012-9571-5>.
- Holzner, B., Wagner, M., 2022. Linking levels of green innovation with profitability under environmental uncertainty: an empirical study. *J. Clean. Prod.* 378, 134438. <https://doi.org/10.3917/jie.pr1.0137>.
- Horbach, J., 2008. Determinants of environmental innovation—New evidence from German panel data sources. *Res. Pol.* 37, 163–173. <https://doi.org/10.1016/j.respol.2007.08.006>.
- Hundloe, T., 2021. Sustainability perspectives: ecological, economic and social. *Environ. Impact Assess* 49–80. https://doi.org/10.1007/978-3-030-80942-3_3.
- IPCC, 2023. Climate change 2023: synthesis report. Contribution of working groups I, II, and III to the sixth assessment report of the intergovernmental Panel on climate change. IPCC, Geneva 35–115. <https://doi.org/10.59327/IPCC/AR6-9789291691647>.
- Jaffe, A.B., 1989. Real effects of academic research. *Am. Econ. Rev.* 79, 957–970.
- Jia, L., Hao, Y., Hu, W., 2021. Energy constraint in China at the expense of profitability? Perspectives of green technology innovation. *J. Global Inf. Manag.* 30, 1–20. <https://doi.org/10.4018/jgim.303110>.
- Khanna, T., Palepu, K.G., 2010. *Winning in Emerging Markets: a Road Map for Strategy and Execution*. Harvard Bus. Press.
- Knott, A.M., Vieregger, C., 2015. *Reconciling the Firm Size and Innovation Debate*. Wharton School, Univ. Pennsylvania, Philadelphia.
- Leal-Rodríguez, A.L., Roldán, J.L., Leal, A.G., Ortega-Gutiérrez, J., 2013. Knowledge management, relational learning, and the effectiveness of innovation outcomes. *Serv. Ind. J.* 33, 1294–1311. <https://doi.org/10.1080/02642069.2013.815735>.
- Le, T.T., 2022. How do corporate social responsibility and green innovation transform corporate green strategy into sustainable firm performance? *J. Clean. Prod.* 362, 132228. <https://doi.org/10.1016/j.jclepro.2022.132228>.

- Leogrande, A., Costantiello, A., 2023. The impact of government expenditure on education in the ESG models at world level. SSRN Working Pap., 4438106 <https://doi.org/10.2139/ssrn.4438106>.
- Leoncini, R., Marzocchi, A., Montresor, S., Rentocchini, F., Rizzo, U., 2019. Better late than never: the interplay between green technology and age for firm growth. *Small Bus. Econ.* 52, 891–904. <https://doi.org/10.1007/s11187-017-9939-6>.
- Leppink, J., Van den Heuvel, A., 2015. The evolution of cognitive load theory and its application to medical education. *Perspect. Med. Educ.* 4, 119–127. <https://doi.org/10.1007/s40037-015-0192-x>.
- Liao, W.W., 2017. A study on the correlations among environmental education, environment-friendly product development, and green innovation capability in an enterprise. *Eurasia J. Math. Sci. Technol. Educ.* 13, 5435–5444. <https://doi.org/10.12973/EURASIA.2017.00841a>.
- Luo, C., Xie, L., 2023. Regional intergenerational mobility and corporate innovation: evidence from China. *PLoS One* 18, e0283588. <https://doi.org/10.1371/journal.pone.0283588>.
- Martínez-Costa, C., Mas-Machuca, M., Benedith content-markup(ce:surname) > , E., Corominas, A., 2014. A review of mathematical programming models for strategic capacity planning in manufacturing. *Int. J. Prod. Econ.* 153, 66–85. <https://doi.org/10.1016/j.ijpe.2014.03.011>.
- Matemilola, B.T., Kijkasiwat, P., Liew, C.Y., 2024. The moderating effect of firm age on capital structure choices: evidence from emerging markets. *J. Ind. Bus. Econ.* <https://doi.org/10.1007/s40812-024-00299-z>.
- Miozzo, M., Soete, L., 2001. Internationalization of services: a technological perspective. *Technol. Forecast. Soc. Change* 67, 159–185. [https://doi.org/10.1016/S0040-1625\(00\)00091-3](https://doi.org/10.1016/S0040-1625(00)00091-3).
- Moulton, B.R., 1990. An illustration of a pitfall in estimating the effects of aggregate variables on micro units. *Rev. Econ. Stat.* 72, 334–338. <https://www.jstor.org/stable/2109724>.
- Nahm, J., 2022. Green growth models. In: Baccaro, L., Blyth, M., Pontusson, J. (Eds.), *Diminishing Returns: the New Politics of Growth and Stagnation*. Oxford Univ. Press, Oxford, pp. 443–464.
- Pinkse, J., 2023. The green transition: net zero as an opportunity to improve productivity. Productivity Institute. Working Pap. No. 024.
- Porter, M.E., Linde, C.V.D., 1995. Toward a new conception of the environment-competitiveness relationship. *J. Econ. Perspect.* 9, 97–118. <https://doi.org/10.1257/jep.9.4.97>.
- Purvis, B., Mao, Y., Robinson, D., 2018. Three pillars of sustainability: in search of conceptual origins. *Sustain. Sci.* 14, 681–695. <https://doi.org/10.1007/s11625-018-0627-5>.
- Putro, R.K., Lubis, A.W., 2023. Inovasi hijau, umur top management, dan profitabilitas: studi pada perusahaan listrik. *Revitalisasi J. Manaj.* 12, 401–411. <https://doi.org/10.32503/revitalisasi.v12i2.4594>.
- Rho, Y., Fabrizi, S., Lippert, S., 2021. Employee Characteristics, Absorptive Capacity and Innovation. University Library of Munich, Germany.
- Riaz, M., Jie, W., Sherani, A., Ali, S., Chang, S., 2024. Assessing the role of organizational strategic factors in stimulating green innovation performance: moderating effects of green absorptive capacity. *Bus. Process Manag. J.* 30, 1013–1043. <https://doi.org/10.1108/bpmj-12-2023-0967>.
- Sala, S., Benini, L., Mancini, L., Ponsioen, T., Laurent, A., van Zelm, R., Pant, R., 2014. Methodology for building LCA-compliant national inventories of emissions and resource extraction. *JRC Sci. Policy Rep.*, EUR, 26871. EN. Publ. Office of the EU, Luxembourg.
- Schweisfurth, T.G., Raasch, C., 2018. Absorptive capacity for need knowledge: antecedents and effects for employee innovativeness. *Res. Pol.* 47, 687–699. <https://doi.org/10.1016/j.respol.2018.01.017>.
- Sen, S., Savitskie, K., Mahto, R.V., Kumar, S., Khanin, D., 2023. Strategic flexibility in small firms. *J. Strat. Market.* 31, 1053–1070. <https://doi.org/10.1080/0965254x.2022.2036223>.
- Shafullah, M., Papavassiliou, V.G., Shahbaz, M., 2021. Is there an extended education-based environmental kuznets curve? An analysis of US states. *Environ. Resour. Econ.* 80, 795–819. <https://doi.org/10.1007/s10640-021-00610-9>.
- Shearmur, R., Doloreux, D., 2016. How open innovation processes vary between urban and remote environments: slow innovators, market-sourced information and frequency of interaction. *Enterpren. Reg. Dev.* 28, 337–357. <https://doi.org/10.1080/08985626.2016.1154984>.
- Shuwaikh, F., Benkraiem, R., Dubocage, E., 2023. Investment in green innovation: how does it contribute to environmental and financial performance? *J. Innov. Econ. Manag.* 41, 107–149. <https://doi.org/10.3917/jie.pr1.0137>.
- Silvério, A., Costa, M., Pereira, L., 2025. Green innovation strategies from the resource-based view perspective. *J. Bus. Strat.* 36, 123–140. <https://doi.org/10.1108/EJIM-12-2024-1427>.
- Stajkovic, A.D., Stajkovic, K.S., 2025. *Human Sustainability and Cognitive Overload at Work: the Psychological Cost of Working*. Taylor & Francis, p. 183.
- Stock, J.H., Yogo, M., 2005. Testing for weak instruments in linear IV regression. In: Andrews, D.W.K., Stock, J.H. (Eds.), *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*. Cambridge Univ. Press, Cambridge, pp. 80–108.
- Stowers, J., 2024. Guide to Choosing a Legal Structure for your Business. *Business News Daily*. <https://www.businessnewsdaily.com/8163-choose-legal-business-structure.html>.
- Teece, D.J., 2016. Dynamic capabilities and entrepreneurial management in large organizations: toward a theory of the (entrepreneurial) firm. *Eur. Econ. Rev.* 100, 86–106. <https://doi.org/10.1016/j.euroecorev.2015.11.006>.
- Wales, W.J., Parida, V., Patel, P.C., 2013. Too much of a good thing? Absorptive capacity, firm performance, and the moderating role of entrepreneurial orientation. *Strateg. Manag. J.* 34, 622–633. <https://doi.org/10.1002/smj.2026>.
- Wang, X., Xu, Z., Qin, Y., Skare, M., 2022. Innovation, the knowledge economy, and green growth: is knowledge-intensive growth really environmentally friendly? *Energy Econ.* 115, 106331. <https://doi.org/10.1016/j.eneco.2022.106331>.
- Wernerfelt, B., 2016. *Adaptation, Specialization, and the Theory of the Firm: Foundations of the Resource-based View*. Cambridge Univ. Press, Cambridge.
- Winston, A., 2024. Why we need to, have to, want to act on climate. *MIT Sloan Manag. Rev.* <https://sloanreview.mit.edu/article/why-we-need-to-have-to-want-to-act-on-climate>.
- Wooldridge, J.M., 2010. *Econometric Analysis of Cross Section and Panel Data*, second ed. MIT Press, Cambridge.
- Yuwono, W., 2021. Empirical analysis of intellectual capital, potential absorptive capacity, realized absorptive capacity and cultural intelligence on innovation. *Manag. Sci. Lett.* 11, 1399–1406.
- Zahra, S.A., George, G., 2002. Absorptive capacity: a review, reconceptualization, and extension. *Acad. Manag. Rev.* 27, 185–203. <https://doi.org/10.2307/4134351>.
- Zhai, T., Chang, Y., 2018. Standing of environmental public-interest litigants in China: evolution, obstacles, and solutions. *J. Environ. Law* 30, 369–397. <https://doi.org/10.1093/jel/eqy011>.
- Zhang, D., 2023. Talk the talk, but walk the walk? female-managed firms and greenwashing behavior. *Environ. Behav.* 55, 364–394. <https://doi.org/10.1177/00139165231192357>.
- Zhang, D., Rong, Z., Ji, Q., 2019. Green innovation and firm performance: evidence from listed companies in China. *Resour. Conserv. Recycl.* 144, 48–55. <https://doi.org/10.1016/j.resconrec.2019.01.023>.